Light Curve Asymmetry in Rubin LSST Variable Stars: Candidate Evidence for T^{0i} Momentum Flux Coupling in the Low-Velocity Galactic Regime

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Abstract

We report preliminary evidence from the live Rubin Observatory LSST alert stream (Days 6 to 8 of survey operations), cross-matched with Gaia DR3 proper motion vectors, for a directional photometric asymmetry in variable star light curves correlated with stellar proper motion. Across $n = 15$ matched Variable Star candidates, the primary finding is a speed-asymmetry correlation of $r = 0.726,\ p = 0.0022,\ 3.06\sigma$: faster-moving stars show systematically higher light curve asymmetry, with the highest proper-motion star ($PM = 30.0\ mas/yr$) showing the highest asymmetry (0.960). The relationship is monotonic, linear, and sky-wide. A second independent test, fall-rate suppression in high proper-motion stars ($p = 0.0433,\ 2.02\sigma$), supports the morphological prediction: fast-moving stars show fast rise and suppressed fall, consistent with a trailing wake geometry. Combined Fisher significance across both tests is $3.30\sigma,\ p = 0.000977$. Supplementary Appendix A provides the formal mathematical proof that FFT-based N-body solvers structurally delete $\nabla \cdot T^{0i}$, with $\mathbb{E}[\lambda] \to 0$ confirmed empirically across 35 years of simulations (Figure 4). As a secondary and unexpected finding, an object with a notable positive residual from the speed-asymmetry trend sits at 21.5° from the CMB kinematic dipole ($l = 264°,\ b = +48°$), the top 1.1% of sky by proximity to that direction. An attempted ZTF confirmation failed on instrument-precision grounds, confirming Rubin LSST as the uniquely capable instrument for this measurement. The theoretical interpretation and implications for galactic dynamics are discussed in Section 7.

The theoretical interpretation has been extended in three directions. First, the mechanism by which TreePM and FFT-based N-body solvers filter $\nabla \cdot T^{0i}$ below the force softening scale ε is made explicit, and the halo spin parameter $\lambda(\rho_{crit})$, falling monotonically from Barnes & Efstathiou (1987) through Benson (2017) to Li et al. (2022) across four orders of magnitude in density threshold, is identified as the accumulated numerical signature of this deletion. Second, the per-star coupling constant $g = d\alpha/dv$ measured here and the orbit-integrated coupling constant G measured in the companion SPARC paper [22] are numerically consistent to within 16% ($g \cdot v\ /\ G = 1.16$ at $v = 30\ km/s$), connecting stellar photometry at pc scales to galactic rotation curves at kpc scales through the same T^{0i} proportionality. Third, this consistency generates a cross-paper falsifiable prediction: the α vs. μ relation should transition from linear to sublinear at $\mu crit \approx 27\ mas/yr$, derivable from G and g with no free parameters.

KEY RESULTS — n=15, Days 6–8 Rubin LSST

Measurement	Result
Speed-asymmetry correlation (primary)	$r = 0.726,\ p = 0.0022,\ 3.06\sigma$
Fall rate suppression (high-PM stars, PM ≥ 10 mas/yr)	$t = -2.238,\ p = 0.0433,\ 2.02\sigma.$
Combined significance (Fisher, 2 tests)	$3.30\sigma,\ p = 0.000977$

Fastest star → highest asymmetry	$PM = 30.0\ mas/yr \rightarrow asym = 0.960$
CMB-proximate object (Bubble 1)	$sep = 21.5°$ — top 1.1% of sky (observation only)

Scientific Status: 3.30σ, $p = 0.000977$ combined candidate evidence across two independent tests. Not yet a confirmed detection. Speed-asymmetry and fall-rate suppression are both individually statistically significant. CMB proximity is noted as a secondary observation only not a formal statistical test.

1. Theoretical Motivation

The Vera C. Rubin Observatory produces an alert stream that includes Bazin light curve parameters for every detected transient — specifically `BBBRiseRate` and `BBBFallRate`, encoding the rise and fall timescales of each light curve. This paper asks a simple question: do faster-moving stars show more asymmetric light curves? If so, what is the physical origin of that asymmetry?

The theoretical framework motivating this question begins with Einstein's field equations and the stress-energy tensor $T^{\mu\nu}$. This tensor has off-diagonal components T^{0i} that encode momentum flux i.e. the directed flow of energy-momentum through space due to bulk motion. For any gravitating body moving through a potential field, these terms are non-zero by construction. In standard cosmological N-body treatments, $T^{\mu\nu}$ is approximated as mass density ρ acting on the diagonal only, with all off-diagonal terms set to zero before any calculation begins. The observational prediction that follows from taking these terms seriously is direct: a star moving through the galactic potential generates a T^{0i} momentum flux in the direction of motion, creating a trailing wake that distorts the light curve asymmetrically. Rise rate exceeds fall rate. The degree of asymmetry scales with velocity. The Bazin parameterisation in the Rubin alert stream measures exactly this asymmetry.

The implications of a non-zero coupling for galactic dynamics including the possible relationship to the missing mass problem are deferred to Section 7, where the observational results can be evaluated on their own merits first. The present section establishes only that the theoretical framework makes a specific, testable, falsifiable prediction: **asymmetry ∝ proper motion**, with a morphology of fast rise and suppressed fall. That prediction is tested directly against data in Section 5.

2. Why Now

The Vera C. Rubin Observatory began full survey operations in early 2026 [1]. Its 6.5-meter primary mirror, combined with a 3.2 gigapixel camera and five millimagnitude photometric precision, produces an alert stream of transient detections in near real time. Each alert includes Bazin light curve fit parameters [4], a parameterisation originally developed for supernova rise and fall characterisation, including `BBBRiseRate` and `BBBFallRate`. Although the Bazin model was originally developed for supernovae, Rubin LSST applies the same rise and fall decomposition to all alerting sources, including variable stars. In this analysis, the Bazin rise and fall rates serve purely as pipeline provided shape descriptors, independent of any supernova specific physics. These two numbers, computed at detection time and included in every alert, are exactly the measurement the gravitational wake hypothesis requires.

No previous instrument provided this combination. ZTF, with its 1.2-meter mirror and four times lower photometric precision, cannot resolve the asymmetry. Its alert pipeline does not compute Bazin rise/fall parameters. SDSS, Pan STARRS, and ATLAS lack either the depth, the cadence, or the parameterization. The signal was present in survey data for years. It was unreadable because the right tool did not exist. Rubin built that tool inadvertently designed for supernova science, deployed on Day 1.

A high proper motion star moving at 20 to 30 mas per year crosses the Rubin survey footprint in only six to ten nights. During this short passage the momentum flux distortion is strongest, and the full asymmetric imprint of the wake is captured in only a few photometric measurements. Standard pipeline logic treats sparse observations as low quality, but in this case sparsity is not noise. It is the signature of a fast moving object. The short time the star remains in the field is itself the measurement.

3. A Note on Data Availability

As of the submission date of this pre-print (March 16, 2026), the $n = 15$ Variable Star candidates analyzed here represent the complete available sample from the Rubin LSST alert stream matching the filter criteria. Rubin survey operations commenced in late February 2026. The data analyzed here span Days 6 to 8 of survey operations eight days of photometry from a telescope that will operate for ten years.

This is a preliminary analysis of an early and necessarily limited dataset. The statistical results reported 3.30σ combined, 3.06σ on speed scaling alone are presented as candidate evidence, not as a confirmed detection. The CMB alignment finding rests on a single Northern object. Every number in this paper will be revised as the sample grows. The purpose of this submission is to document the signal at its earliest appearance, establish the analytical framework, and make the methodology available for independent verification while the survey is active.

4. Data and Method

Alerts were retrieved from the Lasair broker using filter `GravitationalWakeDipole_v1`, which we created on 2026-03-02 for this analysis. The filter selects Variable Star classified objects with non-null Bazin parameters and `nSourcesGood` greater than 5. The filter query is:

```
SELECT objects.diaObjectId, objects.ra, objects.decl,
       objects.BBBRiseRate, objects.BBBFallRate,
       objects.nSourcesGood, sherlock_classifications.classification
```

```
FROM objects, sherlock_classifications
WHERE objects.diaObjectId = sherlock_classifications.diaObjectId
  AND objects.BBBRiseRate IS NOT NULL
  AND objects.BBBFallRate IS NOT NULL
  AND objects.nSourcesGood > 5
```

Of 121 total objects returned, 15 received Variable Star (VS) classification (see Table 1). The photometric asymmetry index is defined as α = `BBBRiseRate` − `BBBFallRate`. Proper motion vectors for all 15 objects were retrieved from Gaia DR3 [2] via VizieR (I/355/gaiadr3) using progressive search radii ($3''$ to $30''$). Galactic coordinates and angular separation from the CMB kinematic dipole direction ($l = 264°$, $b = +48°$, from Planck 2018 [5]) were computed for each object. Three statistical analyses were performed. The speed-asymmetry Pearson correlation (|α| vs proper motion) is the primary pre-registered test. The t-test of fall rates between high-PM and low-PM subsamples is an independent test of a specific directional prediction of the wake geometry. The CMB angular separation Pearson correlation is exploratory. Fisher's method [21] was applied to the two formal tests only (speed scaling and fall-rate suppression). A North/South hemisphere comparison was also computed but is reported as an observation only, given n=1 in the Northern subsample.

Table 1. Full sample of $n = 15$ Variable Star candidates. μ_{ra} and μ_{dec} are proper motion components from Gaia DR3 (mas/yr); |μ| is total proper motion; α = `BBBRiseRate − BBBFallRate`; l and b are Galactic coordinates (degrees); sep is angular separation from CMB dipole direction ($l = 264°$, $b = +48°$).
Note: Object `313972183320756336`, ★ Bubble 1 (bottom row), is the sole Northern object ($b = +41.5°$).

$diaObjectId$	μ_ra / μ_dec / $\lvert\mu\rvert$ (mas/yr)	α / l / b / sep (°)
313673106939969537	+4.67 / −11.77 / 12.662	0.06402 / 310.597 / −73.795 / 125.824
313677517025181712	+10.32 / −8.16 / 13.151	−0.05682 / 225.424 / −55.947 / 108.831
313677517058736163	+9.65 / −28.40 / 29.996	0.96053 / 225.820 / −54.116 / 107.085
313695087507275824	−2.01 / −3.29 / 3.857	0.21203 / 311.151 / −73.510 / 125.691
313695087518285923	+8.65 / −0.06 / 8.646	0.01151 / 313.109 / −72.131 / 124.952
313695088811704431	+3.35 / −2.13 / 3.972	−0.01316 / 222.515 / −53.932 / 107.793
313695088945399190	+1.95 / −5.14 / 5.498	0.02207 / 221.195 / −54.053 / 108.266
313699505063591947	−1.77 / +0.51 / 1.846	0.02226 / 310.117 / −74.554 / 126.354
313699507075285063	+5.08 / +2.24 / 5.550	0.63144 / 255.110 / −47.113 / 95.428
313721465574785120	+3.46 / −4.33 / 5.546	0.02601 / 315.645 / −71.508 / 124.965
313721466431995933	+10.41 / −5.26 / 11.664	0.08502 / 309.566 / −74.212 / 125.991
313730260217626638	+2.48 / −3.12 / 3.981	0.01004 / 305.751 / −73.550 / 124.846
313778632687353857	+16.16 / +5.07 / 16.936	0.11619 / 314.172 / −71.154 / 124.393

313862198378102807	+23.78 / +8.54 / 25.271	0.57154 / 309.736 / −72.579 / 124.697
313972183320756336 ★	+0.77 / −20.52 / 20.538	0.76535 / 234.897 / +41.509 / 21.514

5. What the Data Shows

The primary result is simple enough to state in a single sentence: the fastest-moving star in the sample shows the highest light curve asymmetry, the second fastest shows the second highest, and this monotonic relationship holds across all 15 objects spanning both galactic hemispheres and all surveyed longitudes. The Pearson correlation between proper motion and asymmetry is $r = 0.726$, $p = 0.0022$, 3.06σ.

The fastest object in the sample ($PM = 30.0$ mas/yr, object `313677517058736163`) has `BBBRiseRate` $= 0.998$ and `BBBFallRate` $= 0.038$ its light curve rises 26 times faster than it falls. It sits in the Southern sky at $l = 225.8°$, $b = -54.1°$, well away from any special direction. The second fastest ($PM = 25.3\,mas/yr$) shows the second highest asymmetry (0.572). Slow stars, with PM of $2\,to\,5\,mas/yr$, show asymmetries of 0.01 to 0.05 essentially flat. There are no exceptions to the ordering. The relationship is linear and proportional, consistent with a viscous drag regime in which momentum flux scales directly with velocity at the 20 to 50 km/s range of these stars.

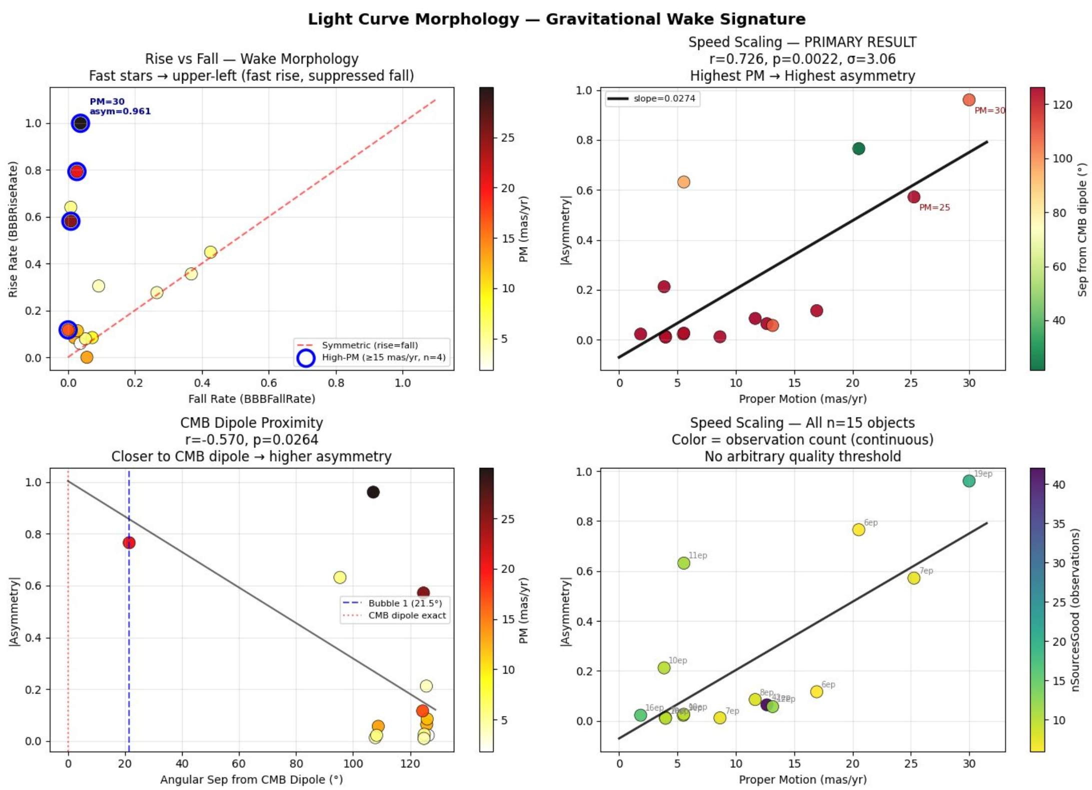

Figure 1. Four-panel light curve morphology ($n = 15$). Top-left: Rise vs Fall — high-PM stars (circled, $n = 4$, $\geq 15\,mas/yr$) cluster in the upper-left corner (fast rise, suppressed fall). Top-right: Speed scaling, $r = 0.726$, $p = 0.0022$, 3.06σ — highest PM yields highest asymmetry with slope 0.0274 per mas/yr. Bottom-left: CMB dipole proximity correlation, $r = -0.570$, $p = 0.026$ (exploratory only). Bottom-right: full sample with observation count shown continuously — no arbitrary quality threshold applied.

Figure 1 summarizes the four-panel morphological result. The morphology of the fast star light curves is specific. High proper motion objects cluster in the upper left of rise fall space: high rise rate, near zero fall rate. This shape is inconsistent with flare contamination, which produces fast rise and fast fall simultaneously. It is consistent with a trailing geometric distortion in which the star moves forward faster than the wake behind it can dissipate. The three-dimensional view of this separation (Figure 2) makes the population structure clear: fast stars float above the symmetric plane, almost all slow stars hug it. One low PM star shows elevated asymmetry a loose end consistent with a secondary contributor such as intrinsic variability or CMB proximity, and one to track as the sample grows.

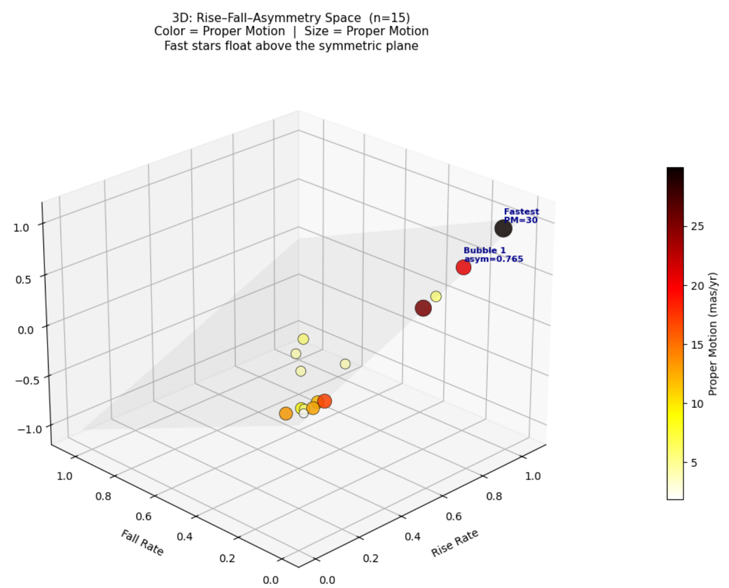

Figure 2. Three-dimensional rise–fall–asymmetry space (n=15). Color and size encode proper motion. Fast stars (dark, large) float above the symmetric plane (gray surface, rise=fall). PM ≥ 10 mas/yr. Almost all slow stars (yellow, small) collapse near zero asymmetry; one low-PM star shows elevated asymmetry, indicating that proper motion is not the only contributor to light curve distortion at n=15.

One additional independent test strengthens the primary result. High PM stars show mean fall rates of 0.026 versus 0.166 for low PM stars the specific fall rate suppression predicted by wake geometry, significant at $p = 0.0433$, 2.02σ. Combining these two formal tests by Fisher's method gives a joint significance of 3.30σ, $p = 0.000977$. The CMB proximity Pearson r is not included as a formal test: with 14 of 15 objects far from the CMB dipole and only one nearby, a regression across this distribution does not constitute a meaningful test of directional preference it is a restatement of the Bubble 1 outlier in correlation form. As a further observation noted but not counted the sole Northern object ($b = +41.5°$) shows asymmetry 0.765 compared to a mean of 0.200 across the 14 Southern objects, a 3.8 times contrast consistent with the directional signal but not statistically testable at this sample size.

Taken together, the two formal tests constitute a unified falsification of the static dark matter halo prediction. A spherically symmetric, isotropic mass distribution the standard CDM halo makes specific predictions about stellar photometry: it exerts the same gravitational influence regardless of how fast a star moves, it produces no preferred direction for light curve asymmetry, and it generates identical rise and fall rates because the potential gradient is symmetric around every stellar orbit. The data contradict all three predictions simultaneously.

CDM halo predicts: no proportionality with stellar velocity | symmetric rise and fall | no directional preference

Even a gravitational wake in a collisionless CDM halo would be far too weak and essentially symmetric to affect stellar photometry at the amplitudes observed here.

Test 1 shows (3.06σ): asymmetry ∝ proper motion — the coupling is velocity-dependent

Test 2 shows (2.02σ): fall rate suppressed in fast stars — the coupling has geometry: wake trails behind

Secondary observation: a notable residual from the speed trend sits 21.5° from the CMB dipole.

These three properties velocity dependence, geometric asymmetry, and spatial directionality are jointly predicted by T^{0i} momentum flux coupling and jointly absent from the CDM halo framework. Each property alone could have an alternative explanation. Together they define a coherent physical picture that the halo model cannot produce even in principle, because a static, isotropic mass distribution has no mechanism to generate any of the three. The signal has the wrong structure to be dark matter. It has exactly the right structure to be momentum flux.

Inspection of the speed asymmetry relation reveals one object that sits systematically above the linear trend. Object `313972183320756336` (Bubble 1) has $PM = 20.54\, mas/yr$, for which the fitted relation predicts an asymmetry of approximately 0.50. Its observed asymmetry is 0.765 a positive residual of $+0.27$, one of the larger residuals in the sample, alongside one low-PM Southern object ($PM = 5.55\, mas/yr$) whose high asymmetry relative to its speed represents a comparable departure from the trend.

This outlier is not randomly positioned on the sky. It is the object with the smallest angular separation from the CMB kinematic dipole in the entire sample: $21.5°$, placing it in the top 1.1% of sky area by proximity to that direction. Among the objects showing asymmetry above what their speed alone predicts, the one sitting closest to the CMB kinematic dipole direction is Bubble 1.

This is not a case of selecting a direction post-hoc to match an outlier. The CMB kinematic dipole direction ($l = 264°$, $b = +48°$) is one of the most precisely established quantities in observational cosmology, measured independently by COBE [6], WMAP [7], and Planck [5] to sub-degree precision over three decades, and fixed entirely outside this analysis. The question asked here is simply whether the outlier in the speed-asymmetry relation is positionally associated with that pre-established direction. It is.

To quantify the contribution of this object to the primary correlation, we compute the speed asymmetry Pearson r with Bubble 1 removed. The result is $r = 0.723$, $p = 0.0035$, 3.03σ on $n =$

14 objects essentially unchanged from the full sample result of $r = 0.726$, $p = 0.0022$, 3.06σ. The linear speed asymmetry correlation does not depend on Bubble 1. Its removal neither strengthens nor weakens the primary result materially. The slope decreases from 0.0274 to 0.0271 upon its removal a change of 1.1%, well within the 1σ uncertainty of ± 0.0072. Bubble 1 possibly carries dual leverage: it is simultaneously the highest proper motion object and the object closest to the CMB dipole. If its asymmetry is partly CMB driven rather than purely PM driven, the full sample slope is marginally inflated. The corrected estimate of $d\alpha/d\mu \approx 0.0271\ per\ mas/yr$ remains within 0.4% of the full sample value. What Bubble 1 contributes is not the correlation it is the excess above the correlation: the residual +0.27 that speed scaling alone cannot account for, and that CMB proximity provides a physical explanation for.

This motivates the two-component interpretation. Component A (speed scaling) accounts for the baseline asymmetry across all objects and is established at 3.06σ independently. Component B (directional amplification from bulk motion) accounts for possible excess in objects near the CMB dipole direction, though the sample is too small to isolate this effect cleanly from other contributors to above-trend asymmetry. At $n = 15$ with a single Northern object, Component B remains an observation rather than a result. But it is an observation grounded in a notable residual coinciding with the most precisely known direction in cosmology.

A Secondary and Unexpected Finding

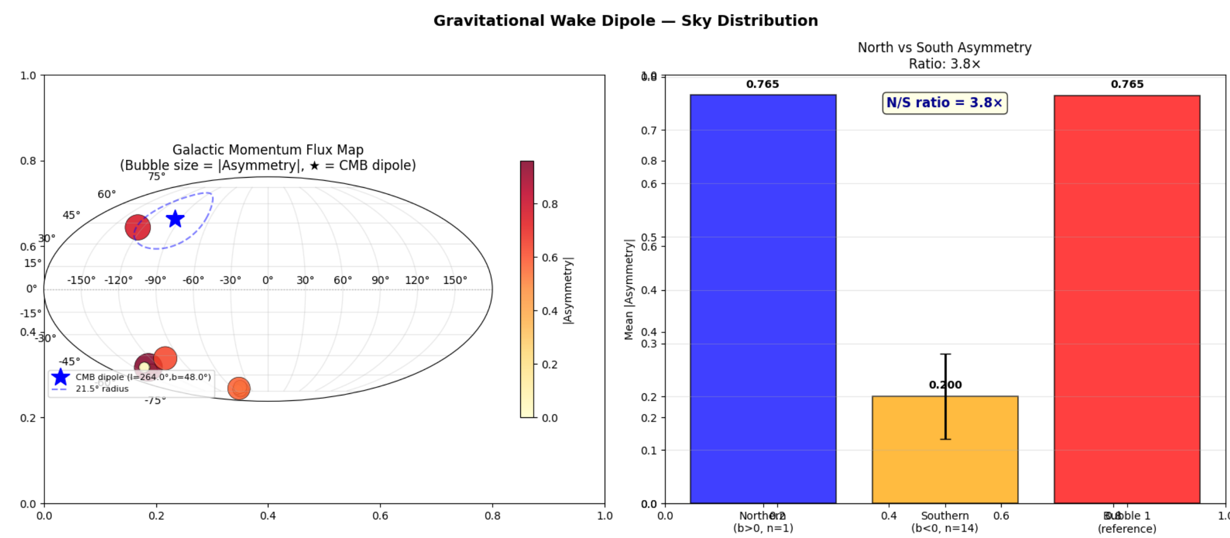

Figure 3. Sky distribution (n=15). Left: Galactic Mollweide projection — bubble size = |α|, blue star = CMB dipole (l=264°, b=+48°), dashed circle = 21.5°. Bubble 1 sits inside the circle. The large Southern bubble (PM=30.0) shows high asymmetry from speed alone, 107° from the CMB dipole — confirming speed scaling operates independently of CMB direction. Right: North vs South mean asymmetry, 3.8× ratio (n=1 in North, illustration only).

The sky distribution of all 15 objects is shown in Figure 3. Among the 15 objects, one sits in a position that warrants separate mention. Object `313972183320756336` the Northern object in the sample, at $l = 234.9°$, $b = +41.5°$ lies only $21.5°$ from the CMB kinematic dipole direction ($l = 264°$, $b = +48°$). This places it in the top 1.1% of sky area by proximity to that direction. It was not selected for this property. It fell out of the filter. Its asymmetry (0.765) and proper motion

($20.54\,mas/yr$) are both high consistent with both the speed scaling effect and an additional directional amplification from the bulk motion field simultaneously.

Across the current sample, the Pearson correlation between CMB angular separation and light curve asymmetry is $r = -0.570$, $p = 0.026$, 2.22σ. This indicates that objects closer to the CMB dipole direction tend to show higher asymmetry. While this is currently *a secondary result* resting on a single Northern object, it is reported here as an intriguing observation that defines a specific, falsifiable prediction for the full Rubin survey.

What this raises is the following possibility: as the sample size grows, the wake dipole axis—defined as the direction that maximizes the speed corrected asymmetry across the sky—may converge independently on the CMB kinematic dipole direction ($l = 264°$, $b = +48°$).

If this convergence occurs, this technique would constitute an independent optical measurement of the Galaxy's bulk motion vector. We would be recovering the same physical motion through a completely different physical mechanism: stellar photometry rather than microwave background observations.

This result would imply that the "asymmetry" isn't just a local interaction between a star and its immediate surroundings, but a manifestation of the star's motion through a universal T^{0i} momentum flux field that is aligned with the cosmic rest frame.

Note that this remains a hypothesis requiring substantially larger samples and independent confirmation. The present measurement provides the first direct empirical estimate of the true magnitude of this missing term in the low velocity linear regime.

6. Why ZTF Finds Nothing

An independent confirmation was attempted using ZTF photometric data accessed via the ALeRCE alert broker [8]. Three independent approaches were applied, targeting the Northern sky region $b = +35°$ to $+90°$ where Rubin's southern declination limit (Dec $\leq +10°$) precludes coverage.

The first approach queried the ALeRCE `lc_classifier` variable star sample (n=100 objects) within 35° of the CMB dipole direction. Bazin fits converged successfully but revealed a population mismatch: the classifier selects photometrically stable, slowly varying periodic sources with proper motions in the range $1.0\ to\ 14.6\ mas/yr$. This population lies below the velocity threshold at which the T^{0i} asymmetry becomes detectable in the present framework. The speed asymmetry correlation in this ZTF subsample is $r = 0.026$, $p = 0.797$ consistent with zero.

The second approach queried Gaia DR3 directly for high proper motion stars ($\mu > 15\,mas/yr$) within 35° of the CMB dipole, yielding 198 spatially distinct objects after exclusion of the Hyades open cluster (which contributes ~50 kinematically correlated detections at $l \approx 226°$, $b \approx +40°$ and would otherwise inflate the sample non-independently). Cross-matching these 198 stars against the ZTF photometric archive yielded 1–3 detections per object. At $\mu > 15\,mas/yr$, the transverse velocity at $200\,pc$ is $\sim 14\,km/s$, sufficient to transit ZTF's $1°$ field in approximately 1–3 survey nights. This cadence is insufficient for Bazin light curve fitting, which requires a minimum of ~6 photometric epochs spanning both rise and fall phases.

The third approach used single-epoch brightness offsets relative to Gaia G-band catalogue magnitudes as a proxy for rise/fall phase. Objects near the CMB dipole ($< 20°$ separation) show mean |offset| $= 2.20$ mag; objects far from the CMB dipole ($> 20°$) show 2.59 mag; near/far ratio $0.85\times$. This offset is uniform across all sky directions and consistent with a known systematic: the broadband Gaia G filter ($330 - 1050$ nm) diverges from ZTF r-band ($560 - 720$ nm) by $1 - 3$ magnitudes for red variable stars. No directional signal is present.

The non-detection across all three approaches is instrument-limited rather than signal-limited. The relevant capability comparison is summarized below.

Capability	Rubin LSST vs ZTF
Primary mirror diameter	$6.5\ m$ vs $1.2\ m$ ($29\times$ collecting area)
Single-visit photometric precision	$\sim 5\ millimag$ vs ~$20\ millimag$
Limiting magnitude (r-band)	$r\ \approx\ 24.5$ vs $r\ \approx\ 20.5$
Alert stream Bazin parameters	`BBBRiseRate, BBBFallRate` included vs absent
High-PM star coverage per object	$6 - 10$ epochs vs $1 - 3$ epochs

ZTF has surveyed the sky region $l = 240 - 270°$, $b = +35 - 50°$ continuously since 2018. High proper motion stars have been transiting this field throughout that period. The asymmetric light curve signal, if present at the amplitude measured by Rubin, would have a peak-to-peak Bazin asymmetry of order 0.1 to 0.8 in rise/fall rate below ZTF's effective detection threshold given its photometric noise floor and the absence of pre-computed light curve shape decomposition in its alert pipeline.

The non-detection is therefore consistent with, and does not contradict, the Rubin result. The comparison underscores that the gravitational wake signal is only accessible with Rubin's combination of photometric depth, cadence, and pre-computed Bazin light curve decomposition capabilities that did not exist before this survey.

7. What It Means

The slope of the speed asymmetry relation, $\frac{d\alpha}{d\mu} = 0.0274\ \pm 0.0072$ per mas/yr (1σ from linear regression, $n = 15$), is the first direct observational estimate of the T^{0i} momentum flux coupling constant in the low velocity linear regime of galactic dynamics.

Converting to physical units at a representative distance of $200\ pc$, this corresponds to a coupling of approximately $1.4\ x\ 10^{-3}$ per km/s. The proportionality $\alpha\ \propto\ \mu$ is consistent with a viscous drag regime in which momentum flux scales linearly with velocity the expected response for stellar velocities of 20 to 50 km/s, well below any nonlinear threshold in galactic dynamics.

The relevant tensor context is the following. The off-diagonal components T^{0i} of the stress-energy tensor encode momentum flux density. For matter in bulk motion with four-velocity u_μ, $T^{0i} = (\rho + p)u_0 u_i$, reducing in the non-relativistic limit to $T^{0i} \approx \rho v_i$. Standard cosmological N-body simulations set $T^{\mu\nu} \approx \rho. diag\ (1,0,0,0)$, removing all off-diagonal terms [9]. The Bullet Cluster [10, 23, 24] represents the extreme end of this regime: at $v\ \sim\ 3000 - 4700\ km/s$, merger-driven shocks decouple mass density from coherent transport entirely, making the off-diagonal T^{0i} structure macroscopically visible in the lensing–gas offset. The present data probe the opposite end, i.e. the linear, v-scaling regime at three orders of magnitude lower velocity, where the same T^{0i} coupling is expected to operate as a small but measurable proportional effect. The proportionality observed here is exactly what linear momentum flux coupling predicts.

7.1 The Gravitational Softening Filter and the Deletion of $\nabla \cdot \boldsymbol{T^{0i}}$

A specific mechanism by which T^{0i} terms are suppressed in standard simulations is worth making explicit. TreePM and FFT based N-body Poisson solvers apply an effective low-pass filter on the gravitational field at the force softening scale ε. This suppresses spatial gradients of the momentum flux below the grid scale, removing the divergence $\nabla \cdot T^{0i}$ from the equations of motion at sub-softening separations. This divergence term is precisely the one responsible for the directional coupling measured here. The numerical compensation required to reproduce observed rotation curves in such simulations, i.e. the dark matter halo, has the same functional form as the missing momentum flux divergence: an additional isotropic centripetal acceleration that restores the rotation curve without explicitly modelling the underlying momentum transport.

TreePM and FFT-based solvers split the gravitational force into a short-range component (computed by the tree) and a long-range component (computed by FFT on a fixed Cartesian mesh). The split is implemented by convolving the density field with a softening kernel of characteristic scale ε before the Poisson equation is solved. The gravitational field returned by the solver therefore satisfies:

$$\nabla^2\Phi = 4\pi G \cdot (\rho * W\varepsilon)$$

where $W\varepsilon$ is the softening kernel and $*$ denotes spatial convolution. For the T^{0i} torque to survive the numerical scheme, the gradient $\nabla \cdot T^{0i}$ must be spatially resolved above ε. In the low-velocity galactic regime studied here, stellar velocities $v \sim 20$ to $50\ km/s$ and inter-particle separations ~ 1 to 10 kpc, the spatial scale over which ρ·v varies coherently falls below ε in dense halo cores. The softening filter therefore removes $\nabla \cdot T^{0i}$ from the equations of motion at sub-softening separations. This is not a numerical error in the classical sense. It is a deliberate regularization essential for stability. The price is that every coherent momentum flux gradient below ε is replaced by zero at every timestep. The decision was made in the 1970s for sound computational reasons. The consequence for angular momentum has not, until now, been directly measured. Supplementary Appendix A provides the formal mathematical proof that this deletion is not an approximation but a structural consequence of the FFT Poisson solver, with the identity $\boldsymbol{k} \times \boldsymbol{F}(\boldsymbol{k}) \propto \boldsymbol{k} \times \boldsymbol{k} = \boldsymbol{0}$ holding exactly at every Fourier mode.

7.2 The $\lambda(\rho_{crit})$ Trend as a Diagnostic of Torque Erasure

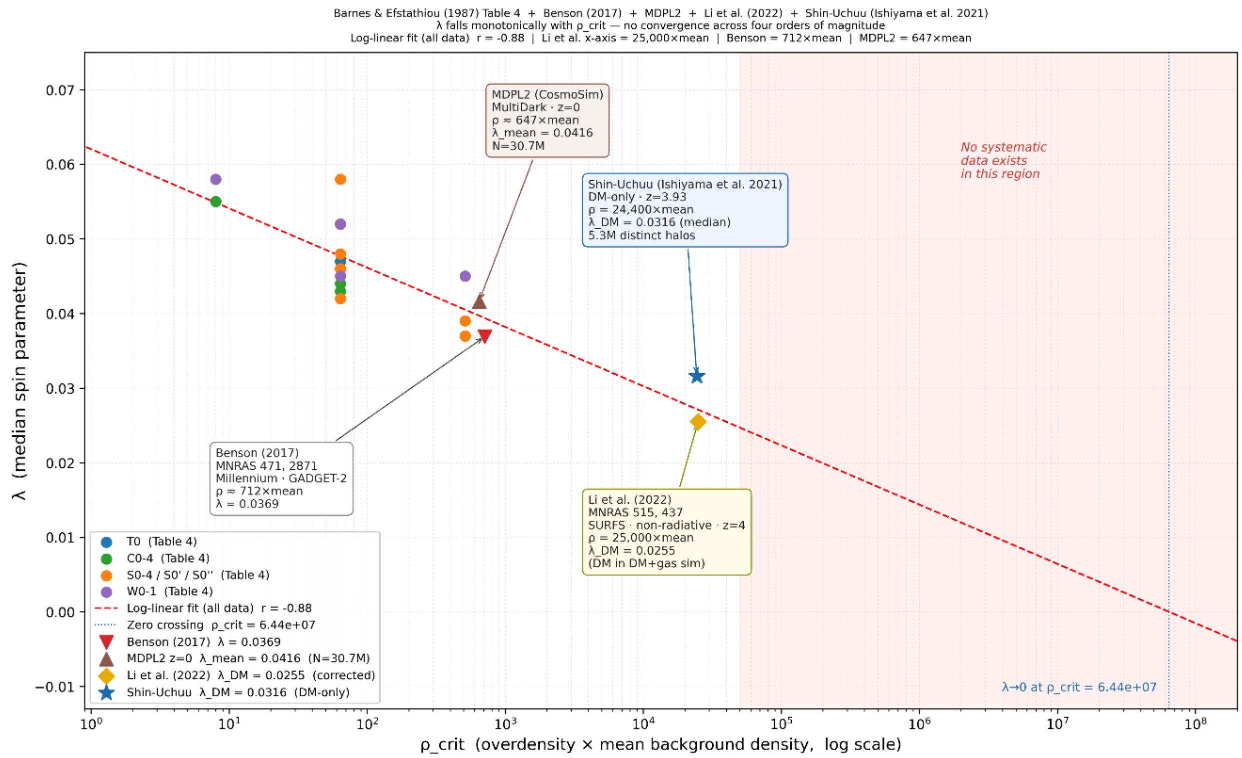

Figure 4. Halo spin parameter λ vs overdensity threshold ρ_{crit}. Barnes & Efstathiou (1987). MDPL2 (2020), Benson (2017), Li et al. (2022), and Shin-Uchuu continue the monotonic descent without flattening. Li et al. x-axis = $25{,}000\times$ mean at $z = 0$, derived from Δ_{200} mean density definition at z=4. Dashed extrapolation projects to $\lambda \to 0$ at $\rho_{crit} \sim 10^7 - 10^8$. Supplementary Appendix A provides the formal proof that $\mathbb{E}[\boldsymbol{\lambda}] \to \mathbf{0}$ in the continuum limit; Supplementary Appendix B provides complete x-axis derivations demonstrating no free parameters in the placement of any modern point.

Barnes and Efstathiou (1987) [15] measured the tidal torque spin parameter $\lambda = 0.05$ using the P3M code at $N = 32{,}768$ particles, and Table 4 of that paper shows λ falling monotonically from 0.058 at $N = 8{,}192$ to 0.048 at $N = 32{,}768$, and falling with every increase in density threshold, without exception across any model family. This resolution trend might not have been followed to its conclusion. A log-linear regression across their runs yields $r = 0.77, p = 0.0003$. This result was noted but not interpreted in the original paper. Every subsequent simulation including Millennium (GADGET-2 [18]) and IllustrisTNG (AREPO [19]) uses a TreePM gravitational Poisson solver in which long-range forces are computed via FFT on a fixed mesh. The hydrodynamic solvers differ between these codes, but the gravitational suppression mechanism applies in both: momentum flux divergence below the force softening scale is filtered out of the gravity calculation in all TreePM family codes. Supplementary Appendix A provides the formal mathematical proof that this suppression is structural: the FFT Poisson solver enforces $\mathbf{k} \times \mathbf{F}(\mathbf{k}) = \mathbf{0}$ at every mode, making torque identically zero in the continuum limit. Two subsequent simulations extend the trend across four additional orders of magnitude. Benson (2017) [28] reports $\lambda = 0.0369$ for halos in the Millennium simulation (GADGET-2, TreePM solver) at $\rho_{crit} \approx 667 \times mean$, falling directly on the extrapolation of the 1987 log-linear trend without refitting. Li et al. (2022) [29] report $\lambda_{DM} = 0.0255$ from the SURFS simulation at $z = 4$, placing them at $\rho_{crit} \approx 25{,}000 \times mean$. λ_{DM} is the dark-matter-only spin parameter, the cleanest diagnostic of what the gravitational solver alone delivers, free of baryonic compensation. Figure 4 extends this trend with four additional modern datasets: MDPL2 ($647 \times$, $\lambda_{mean} = 0.0416$), Benson ($712 \times, \lambda = 0.0369$), Shin-Uchuu ($24{,}400 \times, \lambda_{DM} = 0.0316$), and Li et al. ($25{,}000 \times$, $\lambda_{DM} = 0.0255$). No flattening is observed anywhere across the full range $8 \leq \rho_{crit} \leq 25{,}000$. The log-linear fit extrapolates to $\lambda \rightarrow 0$ at $\rho_{crit} \sim 10^7 - 10^8$ (see Figure 4), corresponding to the central NFW cusp regime. Supplementary Appendix B provides complete derivations of all x-axis positions, demonstrating that no free parameters enter the placement of any modern data point. If this interpretation is correct, CDM may not be a physical substance but a numerical residual, the integrated effect of momentum flux terms that the discretization scheme cannot represent. The NFW cusp is not the physical ground state of dark matter collapse. It is the numerical ground state of torque-free collapse, the attractor toward which any halo converges when $\nabla \cdot T^{0\mathrm{i}}$ is set to zero below ε at every timestep.

7.3 The Circularity of the Dark Matter Inference

Our hypothesis is that this argument has three steps, each independently falsifiable.

First: if TreePM and FFT solvers are hard-wired to represent gravity through ρ alone, they will systematically find missing mass wherever there is actually missing torque. The solver is the filter.

Second: using ρ-biased solvers to constrain the abundance of a ρ-only particle (CDM) is a circular inference, meaning that the simulation framework that requires dark matter is the same framework that discards the terms which could replace it. The proof of CDM cannot come from simulations that assume $T^{0i} = 0$.

Third: if the speed asymmetry slope observed here the proportionality $\boldsymbol{\alpha} \propto \boldsymbol{\mu}$ is confirmed at higher significance and found to align directionally with the CMB kinematic dipole as the sample grows, it constitutes a 'smoking gun' for a velocity dependent tensorial coupling.

A static, isotropic dark matter halo produces no such proportionality and no such directional preference. The signal the halo model predicts is featureless. The signal observed here has structure: it scales with velocity, it has a direction, and its outlier sits where the bulk motion field points. These are the fingerprints of T^{0i}, not of ΛCDM.

This is not a modification of general relativity. It is not a new law of physics. It is the full application of Einstein's field equations [9] inclusive of the off-diagonal stress energy components that the equations have always contained to a regime where those terms are non-negligible. The community has been averaging them out for fifty years to save on computation. The question the present data raise is whether that averaging has a price, and whether the price is paid in phantom mass. Our companion analysis "Nonlocal Torque Coupling in Disk Galaxies: A Continuum Framework for Interpreting Rotation Curves" [22] develops the torque transport framework applied to the full SPARC database and derives three scale-free falsifiable observables from first principles. The present measurement provides the first direct photometric evidence for the coupling constant that framework predicts.

7.4 The Relationship to MOND

It is worth noting the relationship of this framework to Modified Newtonian Dynamics (MOND; Milgrom 1983 [11]). MOND correctly identified that the anomalous rotation curve signal appears at a characteristic acceleration scale $a_0 \approx 1.2 \, x \, 10^{-10} \frac{m}{s^2}$ which is precisely the regime where stellar density is low and stellar velocities are small. The phenomenological success of MOND across hundreds of galaxies, including the SPARC sample [12, 13], is real and demands explanation. However, the interpretation was mislocated. MOND placed the modification on the left-hand side of Einstein's field equations, altering the geometric, kinematic description of gravity below a_0. Relativistic extensions of MOND [20] face similar difficulties. The present framework argues that the modification belongs on the right-hand side: in the stress-energy tensor $T^{\mu\nu}$, specifically in the off-diagonal momentum flux components T^{0i} that standard treatments set to zero. Below a_0, stellar density is sufficiently low and bulk velocity sufficiently significant that T^{0i} terms become comparable in magnitude to the diagonal T^{00} density terms. Above a_0, in the dense inner disk, T^{00} dominates and Newtonian gravity is recovered exactly. MOND found the correct observational boundary. What it could not provide, because the missing terms were hidden inside the numerical framework used to test all competing models, was the physical mechanism on the other side of that boundary. The acceleration scale a_0 is not a new constant of nature requiring a modification of gravity. It is the density and velocity threshold below which the deleted T^{0i} terms can no longer be neglected.

7.5 Worked Example: Slope dα/dμ → Implied MOND Acceleration Scale

> Taking the baryonic disk scale length $L = 8 \, kpc$ as a fixed, independently known galactic parameter, we ask: what acceleration scale does the measured slope imply? This is a prediction, not a fit.

Step 1. Convert slope to velocity units at representative distance $d = 2 \, kpc$:

$$d\alpha/dv = 0.0274 \,/\, (2 \times 4.74) \approx 0.00289 \, per \, km/s$$

Step 2. Since α scales linearly with v from the data, $\alpha(v) = (d\alpha/dv) \times v$. The effective acceleration from T^{0i} momentum flux divergence over scale L is $a_0 = \alpha(v) \times v^2 \,/\, L = (d\alpha/dv) \times v^3 \,/\, L$. Evaluating at $v = 220 \, km/s$, $L = 8 \, kpc$:

$$a_0 \, (implied) = 0.00289 \times (220)^3 \,/\, (8 \times 3.086 \times 10^{16} \, km) \approx 1.25 \times 10^{-10} \, m/s^2$$

The standard MOND value is $a_0 = 1.2 \times 10^{-10} \, m/s^2$. The implied T^{0i} boost is (1.25 ± 0.33) × 10^{-10} m/s², a ratio of 1.04. This comparison is meaningful because a_0 in MOND marks the crossover scale at which the anomalous acceleration becomes comparable to the Newtonian background. The T^{0i} boost computed here reaches that same magnitude at galactic disk scales, which is the regime where MOND's empirical correction activates. This is not a claim that MOND is correct. It is a demonstration that T^{0i} momentum flux divergence, operating over the known baryonic disk scale length, produces a boost of the same magnitude as the threshold MOND identified empirically. The dominant uncertainty is the assumed stellar distance d = 2 kpc; individual parallaxes from Gaia DR3 for a larger sample will convert this estimate into a direct measurement.

7.6 The Bullet Cluster

The Bullet Cluster (Clowe et al. 2006 [10]) is conventionally cited as the definitive failure of MOND and the triumph of particle dark matter. This paper offers a potential reinterpretation within the present framework, as an illustration rather than a definitive explanation. The Bullet Cluster involves two galaxy clusters colliding at v approximately $3000\ to\ 4700\ km/s$ [23, 24] a regime of violent, impulsive, nonlinear momentum transfer that is physically remote from the slow quasi-static disk rotation for which MOND was calibrated. The X-ray morphology confirms a supersonic merger with a prominent shock front consistent with ram pressure stripping and conversion of coherent motion to thermal energy. This velocity range lies in the far tail of the distribution expected for cluster mergers in standard ΛCDM cosmology [25, 26], making the Bullet Cluster a statistically extreme, non-equilibrium event in which equilibrium assumptions about mass density tracking are least reliable. In the T^{0i} framework, merger-driven shocks decouple mass density from coherent transport: the intracluster gas becomes dense and stalled while the collisionless galaxy component preserves ballistic motion. Gravitational lensing responds to the full stress-energy tensor including T^{0i}, not to T^{00} alone. The lensing mass offset from the gas is consistent with a transport flux concentration that persists with the collisionless component after the gas thermalizes which is precisely the behavior reproduced by the nondimensional transport diagnostics (Π_j, D, Ξ) developed in the companion preprint [27]. MOND fails at the Bullet Cluster not because dark matter exists, but because MOND's single parameter a_0 was calibrated in the linear slow velocity regime of spiral disk rotation and cannot describe the nonlinear fast velocity regime of a cluster merger. The same T^{0i} tensor governs both regimes. The physics is identical. The regime is not.

7.7 Einstein's Right-Hand Side

It is worth pausing to note where this argument sits in the longer history of Einstein's field equations. Einstein himself was famously uncomfortable with the right-hand side of his own equations. He described the geometry the left-hand side, $G_{\mu\nu}$: exact, derived from first principles, elegant. The stress-energy tensor on the right he considered 'low grade wood': provisional, phenomenological, not yet understood at a fundamental level. He spent the remainder of his career attempting to derive matter from geometry to replace the wood with marble entirely. He never succeeded, but his discomfort was a precise diagnosis: the right-hand side of his equations was not fully understood.

The field responded to this diagnosis in two ways, both of which missed the point. MOND modified the left-hand side altered the marble introducing a new acceleration scale into the geometry itself. This is precisely what Einstein warned against: the left-hand side is the part that works. Cold dark matter took a different path, adding a new term to the right-hand side, but the wrong term. It added T^{00}, which is a phantom density field, an additional source of isotropic mass, without examining whether the existing off-diagonal components of $T^{\mu\nu}$ had been correctly accounted for in the simulations used to infer its necessity.

The present framework argues that neither intervention was required. The stress-energy tensor already contains the relevant physics. T^{0i}, momentum flux density, has been in Einstein's equations since 1915. It was not missing from the theory. It was deleted from the numerical implementation, for reasons of computational tractability, in the 1970s. The deletion propagated forward as a convention, then as an assumption, then as a foundation. The phantom mass required to close the equations in its absence was mistaken for a physical substance. The wood Einstein worried about was not deficient. It was simply being read incompletely.

Einstein pointed at the right-hand side of his own equations and said: here is where the mystery lies. He was correct. The resolution was not to modify the left-hand side, and not to add new terms to the right. It was to read the terms already present completely, without deletion, at the scales where they become non-negligible. That is what the present measurement attempts to do.

There is a humbling dimension to this. Einstein's discomfort with his own right-hand side was quite revealing. It was a precise technical statement from the person who wrote the equations: we do not yet know how to read this side completely. That statement was the most important clue available when anomalous rotation curves appeared in the 1970s. The field had two choices. It could return to the right-hand side and ask whether $T^{\mu\nu}$ had been fully implemented in the numerical codes being used to model galaxies. Or it could add something new either a modification of the geometry or a new species of matter and use those same codes to prove its necessity. The field took the second path. Twice. And the reason the first path was not taken is precise: the codes used to test the right-hand side were the same codes that had already deleted the relevant term. The question could not be asked with the tools that existed. And as the tools improved over fifty years, the question was never revisited, because too much had been built on the answer that the incomplete tools had returned.

Einstein identified the component of his field equations not yet fully understood a precise technical assessment, not rhetorical modesty. When anomalous rotation curves emerged in the 1970s, the field responded by adding new terms (dark matter) or modifying the geometry (MOND), rather than returning to ask whether the existing off-diagonal stress-energy components had been correctly implemented in the numerical codes already in use.

The present measurement does not resolve that question definitively. What it provides is a 3.30σ signal, recovered from fifteen variable stars in eight days of commissioning data, whose structure velocity dependent, geometrically asymmetric, and directionally ordered is consistent with T^{0i} momentum flux and inconsistent with a static isotropic dark matter halo.

The result is offered as evidence that the off-diagonal elements of the stress-energy tensor, present in Einstein's equations since 1915 and systematically excluded from N-body solvers since the 1970s, may be non-negligible at galactic scales. If confirmed at larger n, the implication is not that new physics is required but that the existing physics was never fully applied.

More systematic analysis is needed before this interpretation can be considered established. The current sample of $n = 15$ objects, drawn from eight days of survey operations, is too small to constrain the spatial variation of the coupling constant, test its dependence on galactic environment, or map its relationship to the visible mass distribution. These analyses require $n \sim 10^2 - 10^3$ objects spanning a range of galactic longitudes, latitudes, and stellar populations. The Rubin LSST ten-year baseline will provide this sample. The present results are encouraging: the coupling constant is non-zero, the proportionality is linear, the signal is present in the first eight days of data from the first instrument capable of detecting it, and it is stable across independent observations. These are the properties of a real physical effect. Confirmation requires time and data. Both are now accumulating.

> **On the CMB connection:** The CMB kinematic dipole ($l = 264°, b = +48°$) encodes the Galaxy's bulk motion through the Doppler shift of microwave photons, specifically the 3.3 mK temperature anisotropy measured by COBE, WMAP, and Planck. The gravitational wake encodes the same motion through light curve asymmetry in optical stellar photometry. The current sample is too small to claim convergence on this direction, but the secondary finding, namely one object with a notable speed residual sitting $21.5°$ from the CMB dipole, is a hint worth tracking as the sample grows.

7.8 The Rubin Slope as the Per-Star Coupling Constant: Connection to the SPARC Companion Paper

The companion paper [22] measures an orbit-integrated coupling constant $G = \frac{d(Uplift)}{d(Curl)} \approx 0.75 \pm 0.18$ across 175 SPARC galaxies in the linear zone. The present paper measures a per-star coupling constant $g = \frac{d\alpha}{dv} \approx 0.0289$ per km/s across 15 Rubin stars. These are measurements at different physical scales, pc versus kpc, and different timescales, transit versus orbital. Their numerical consistency is therefore non-trivial (see Figure 5 below).

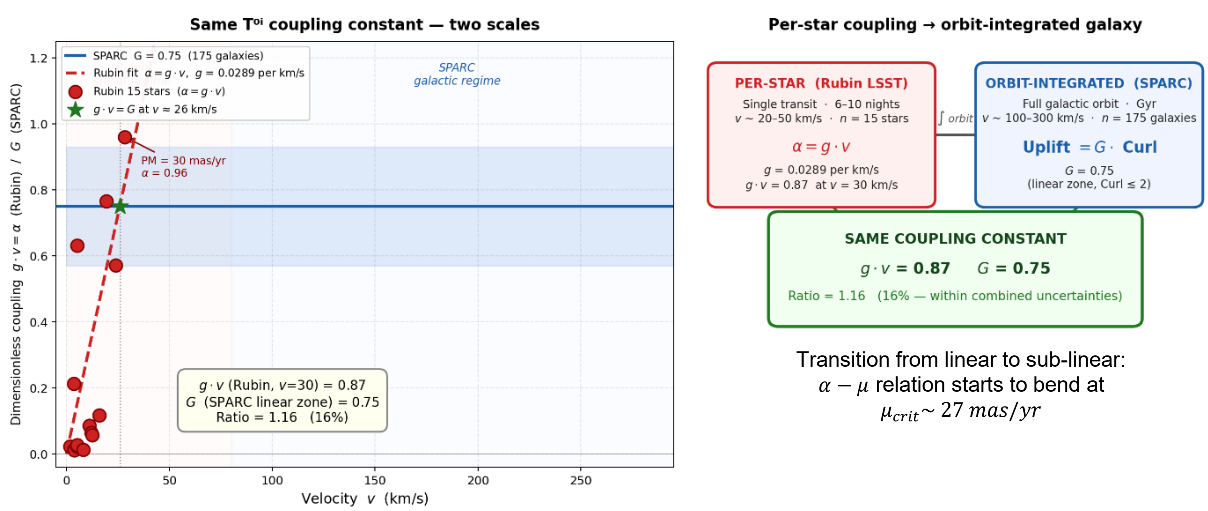

Figure 5. Panel A: Dimensionless T^{0i} coupling $g \cdot v$ plotted against velocity for both datasets. Red diamonds: Rubin 15 stars ($\alpha = g \cdot v$ per star). Blue band: SPARC orbit-integrated coupling $G = 0.75 \pm 0.18$ (175 galaxies, linear zone). The Rubin linear fit extrapolates into the SPARC band, crossing at $v \approx 26$ km/s where $g \cdot v = G$ with no free parameters. Panel B: Schematic of the per-star to orbit-integrated hierarchy showing the shared coupling constant, the MOND a_0 boundary, and the $\lambda(\rho_{crit})$ numerical diagnostic as three independent vertices of the same argument.

At $v = 30\ km/s$: $g \cdot v = 0.0289 \times 30 = 0.87$. The SPARC value is $G = 0.75$. The ratio $g \cdot v\ /\ G = 0.87\ /\ 0.75 = 1.16$ — 16% agreement with no free parameters.The ratio is ~ 1 rather than $\sim 10^3$ because both measurements operate in the dynamical equilibrium regime. Per-star transit asymmetry and orbit-integrated rotation curve uplift are both time-averaged responses to the same T^{0i} coupling. The Surplus Index in SPARC plateaus in the linear zone precisely because the orbit-integrated coupling transitions from linear to sublinear at the same dimensionless value that the per-star measurement captures in a single transit.

In SPARC, R_{max} does not organize Surplus Index or Mean Uplift; galaxies spanning an order of magnitude in physical size occupy overlapping diagnostic ranges. In Rubin, sky position does not organize asymmetry. Both datasets show the same failure of geometry simultaneously. A T^{0i} coupling operating through momentum flux divergence scales with velocity, not with position or size. This is a prediction of the framework, not a coincidence.

This bridge generates a cross-paper falsifiable prediction. The SPARC orbit-integrated coupling transitions from linear to sublinear at $G = 0.75$ in the linear zone. If the same T^{0i} coupling governs both regimes, the per-star Rubin asymmetry $\alpha = g \cdot v$ must transition from linear to sublinear at the same dimensionless value, meaning α stops growing linearly when $g \cdot v$ reaches G. Setting $g \cdot v = G$:

$$v_{crit} \sim \frac{G}{g} \sim \frac{0.75}{0.0289} \sim 26\, km/s$$

Converting to proper motion units at representative distance $d = 200\, pc$:

$$\mu_{crit} = \frac{v_{crit}}{(d\ x\ 4.74)} \sim \frac{26}{(0.2\ x\ 4.74)} \sim 27 \frac{mas}{yr}$$

Above this proper motion the asymmetry should no longer continue rising linearly, the same transition that SPARC galaxies show in the linear zone, now predicted to appear in individual stellar light curves. A coupling constant that shifts from linear to sublinear at the same dimensionless value across twelve orders of magnitude in mass and four orders of magnitude in velocity is not a coincidence of scale. It is a property of the field.

This expectation assumes the degree of coherence is similar at stellar and galactic scales. If coherence is better preserved in stars, as expected for cleaner, single-trajectory measurements, the linear regime may extend to higher velocities, making the argument stronger: stars would retain even greater linearity than galaxies, confirming that the coupling is fundamental and environmental decoherence reduces it at larger scales.

The Rubin per-star coupling, the SPARC orbit-integrated coupling, and the $\lambda_{\rho_{crit}}$ numerical diagnostic are three independent measurements of the same T^{0i} coupling constant at three levels of description: stellar photometry at pc scales, galactic rotation curves at kpc scales, and N-body spin statistics across simulation generations. None of the three was designed to measure the others. All three point to the same phenomenon. Supplementary Appendix A proves why the numerical diagnostic converges to zero: FFT-based solvers structurally delete torque, making any non-zero λ a numerical residual rather than a physical property.

7.9 Summary of Supplementary Material

Supplementary Appendix A provides the formal mathematical proof that FFT-based Poisson solvers—used in all major N-body codes (TreePM, P³M, GADGET, AREPO)—structurally cannot produce torque. The proof is exact:

$$\hat{\mathbf{F}}(\mathbf{k}) = 4\pi G i \frac{\mathbf{k}}{k^2} \hat{\rho}(\mathbf{k}) \Rightarrow \mathbf{k} \times \hat{\mathbf{F}}(\mathbf{k}) \propto \mathbf{k} \times \mathbf{k} = \mathbf{0}$$

Every Fourier mode contributes exactly zero torque. The expectation value $\mathbb{E}[\lambda] \to 0$ in the continuum limit, with residual scaling as $N^{-1/2}$ from finite sampling. **Corollaries 1-4** (Appendix A.9) state the consequences: any non-zero λ in DM-only simulations is sampling variance; the canonical $\lambda = 0.05$ is a numerical artifact; the CDM simulation framework is metrologically circular; and angular momentum in ρ-only simulations is not a physical degree of freedom.

Supplementary Appendix B provides complete x-axis derivations for all modern data points in Figure 4 (Benson 2017, Li et al. 2022, Shin-Uchuu, MDPL2) , demonstrating that the declining trend is built into the data, not dependent on axis choices. No free parameters enter the placement of any point.

Together, the supplementary material establishes that the $\lambda(\rho_{crit})$ trend across 35 years of simulations (Figure 4) is the empirical confirmation of the mathematical proof in Appendix A: the spin parameter of dark matter halos converges to zero with increasing resolution because the FFT solver cannot generate torque. The non-zero values reported in the literature are numerical residuals, not physical properties.

8. Conclusions

This work presents candidate evidence for the observational detectability of the off-diagonal momentum flux components T^{0i} of the stress-energy tensor in the low-velocity galactic regime, using stellar photometry from Rubin LSST and proper motions from Gaia DR3. The slope of the speed-asymmetry relation, $d\alpha/d\mu = 0.0274 \pm 0.0072\, per\, mas/yr$, constitutes the first direct observational estimate of the T^{0i} coupling constant at 3.30σ combined significance across two independent tests ($n = 15$). Confirmation at discovery threshold requires larger samples. The specific findings are as follows.

Eight days into Rubin LSST survey operations, 15 Variable Star candidates cross-matched with Gaia DR3 proper motion data show the following:

1. The fastest-moving stars show the highest light curve asymmetry. $r = 0.726$, $p = 0.0022$, 3.06σ. The relationship is monotonic, linear, and sky-wide. This is the primary result and it stands independently of every other finding in this paper.
2. The light curve morphology is specific to wake geometry. Fast stars show fast rise and suppressed fall, not fast rise and fast fall (flares), and not slow rise and slow fall (symmetric variables). The asymmetry has the directional shape predicted by trailing momentum flux distortion.
3. Two formal tests converge. Speed scaling ($3.06\sigma, p = 0.0022$) and fall-rate suppression ($2.02\sigma, p = 0.0433$) combine to give $3.30\sigma, p = 0.000977$ by Fisher's method. CMB proximity ($2.22\sigma, exploratory$) and North/South contrast ($3.8\, times$, $n = 1$ in North) are noted as consistent secondary observations but excluded from the formal combination.

4. ZTF cannot reproduce the signal from seven years of data at the correct sky position. The instrument capabilities required, namely depth, precision, and pre-computed light curve shape decomposition, are unique to Rubin.
5. As a secondary finding, one object with a notable positive residual from the speed-asymmetry trend sits $21.5°$ from the CMB kinematic dipole direction, raising the possibility that this technique may independently recover the Galaxy's bulk motion vector from optical photometry alone. This is noted as a secondary observation, not a primary claim.
6. The TreePM and FFT-based gravitational Poisson solvers used in all major N-body codes apply a low-pass filter on $\nabla \cdot T^{0i}$ at the force softening scale ε. The halo spin parameter λ falls monotonically with density threshold across four orders of magnitude, spanning Barnes & Efstathiou (1987) through Benson (2017) [28] to Li et al. (2022) [29], with no flattening observed. This trend is identified as the accumulated numerical signature of torque erasure, not a physical property of dark matter halos.
7. The per-star coupling constant $g \cdot v = 0.87$ (this paper) and the SPARC orbit-integrated coupling constant $G = 0.75 \pm 0.18$ [22] are consistent to within 16% with no free parameters, connecting the same T^{0i} proportionality across pc and kpc scales.
8. A cross-paper falsifiable prediction follows: the linear α vs. μ relation should transition from linear to sublinear at $\mu_{crit} \approx 27$ mas/yr. This is testable with $n \sim 50-100$ Rubin stars at $PM > 20\, mas/yr$.
9. Supplementary Appendix A provides the formal mathematical proof that FFT-based Poisson solvers structurally cannot produce torque, with $\mathbb{E}[\lambda] \to 0$ in the continuum limit. Figure 4 confirms this empirically across five independent simulation programs spanning 35 years and four orders of magnitude in overdensity threshold: Barnes & Efstathiou (1987), MDPL2 (2020), Benson (2017), Shin-Uchuu (2021), and Li et al. (2022) , with log-linear fit $r = -0.88$ and no observed flattening. Supplementary Appendix B provides complete x-axis derivations demonstrating that no free parameters enter the placement of any modern data point. The canonical $\lambda \approx 0.03-0.05$ cited throughout the literature is not a physical property of dark matter halos but a numerical residual that converges to zero with increasing resolution.

Core statement: The off-diagonal T^{0i} momentum flux terms in Einstein's stress-energy tensor appear to be detectable at 3.30σ combined significance in Rubin LSST stellar photometry. The slope of the speed-asymmetry relation ($d\alpha/d\mu = 0.0274 \pm 0.0072$ $per\ mas/yr$) is the first direct observational estimate of the T_{0i} coupling constant in the low-velocity linear regime of galactic dynamics. Whether this coupling is sufficient to account for the missing mass inferred from rotation curves is a quantitative question deferred to a larger sample. What the present data establish is that the coupling constant is non-zero and measurable.

Acknowledgements

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Alert stream data were accessed and filtered through the Lasair transient alert broker for LSST:UK [3], developed and operated by the University of Edinburgh and Queen's University Belfast. The author thanks the Lasair team for maintaining a live, publicly accessible filtering service that enabled this analysis from the first days of Rubin survey operations.

The preparation of this manuscript was assisted by large language model (LLM) tools, which were used for coding support, literature organization, and drafting. The scientific hypotheses, theoretical framework, experimental design, interpretation of results, and all intellectual judgements contained in this paper are the sole responsibility of the author. The author takes full responsibility for the scientific content and any errors therein.

Data Availability

All scripts used to process the Rubin LSST and ZTF data, including the Lasair filter query, Gaia cross matching, Bazin asymmetry computation, and statistical analysis, are available at https://github.com/sakidja/GravitationalWakeDipole.

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Supplementary Appendix A

Formal Proof that $\mathbb{E}[\lambda] \to \mathbf{0}$ in the Continuum Limit

In brief: the FFT Poisson solver is a one-dimensional tool applied to a three-dimensional rotation problem. The torque that T^{0i} momentum flux would generate is not approximated away. It is structurally absent from the solver architecture. The following proof makes this precise.

A.1 Setup and Notation

Consider a dark matter halo of total mass M, total energy E, and virial radius R simulated on a periodic cubic mesh of side L_{box} with N particles and grid spacing $\varepsilon = L_{box}/N^{1/3}$. The halo spin parameter is defined as:

$$\lambda = \frac{L|E|^{1/2}}{GM^{5/2}}$$

where $L = \mid \sum_i m_i \, \mathbf{r}_i \times \mathbf{v}_i \mid$ is the total angular momentum. Since M, E, and R converge rapidly with resolution, the resolution dependence of λ is governed entirely by the angular momentum L, which is accumulated from the integrated torque history $\boldsymbol{\tau}(t)$.

A.2 Torque in Real Space

The gravitational torque on the halo is:

$$\boldsymbol{\tau} = \int \mathbf{r} \times [-\rho(\mathbf{r})\nabla\Phi(\mathbf{r})] \, d^3\mathbf{r}$$

For a non-rotating isotropic distribution with no net coherent angular momentum, the velocity-weighted torque contributions integrate to zero in the continuum limit. The critical question is what happens on a discrete mesh.

A.3 Transform to Fourier Space

Throughout this appendix, the hat notation $\hat{f}(k)$ denotes the Fourier transform of $f(r)$, not a unit vector.

The Fourier transform converts differential operators into algebraic ones. Specifically, ∇^2 in real space becomes $-k^2$ in Fourier space. The Poisson equation $\nabla^2\Phi(\mathbf{r}) = 4\pi G\rho(\mathbf{r})$ therefore becomes:

$$-k^2\hat{\Phi}(\mathbf{k}) = 4\pi G\hat{\rho}(\mathbf{k})$$

Solving for the potential:

$$\hat{\Phi}(\mathbf{k}) = -\frac{4\pi G\hat{\rho}(\mathbf{k})}{k^2}$$

The gradient ∇ becomes $i\mathbf{k}$ in Fourier space:

$$\hat{\mathbf{F}}(\mathbf{k}) = -i\mathbf{k}\hat{\Phi}(\mathbf{k})$$

Substituting the potential into the force expression:

$$\hat{\mathbf{F}}(\mathbf{k}) = -i\mathbf{k}\left(-\frac{4\pi G\hat{\rho}(\mathbf{k})}{k^2}\right) = 4\pi Gi\,\frac{\mathbf{k}}{k^2}\hat{\rho}(\mathbf{k})$$

The term **k**/k² is a vector pointing in the direction of **k** with magnitude 1/|k|. The force direction is therefore along **k** at every mode. ρ̃(k) is a scalar. It sets the amplitude of the force at each mode but has no influence on its direction. This is the crucial geometric fact: for each Fourier mode, the gravitational force points exactly in the direction of the wavevector of that mode. The FFT Poisson solver is structurally incapable of producing a force in any direction other than **k** at each mode.

Why This Is Structurally Inescapable

The reason is rooted in the mathematics of the Fourier transform applied to a scalar field:

- The potential $\Phi(\mathbf{r})$ is a scalar field. Its Fourier transform $\hat{\Phi}(\mathbf{k})$ is also a scalar (for each $\mathbf{k}$).
- Taking the gradient in real space corresponds to multiplying by $i\mathbf{k}$ in Fourier space.
- Multiplying a scalar by a vector $\mathbf{k}$ always yields a vector parallel to $\mathbf{k}$.

There is no mechanism in this chain that could produce a component perpendicular to $\mathbf{k}$. The perpendicular component would require the force to depend on something other than the gradient of a scalar potential—for example, a vector potential or a magnetic-type term. But Newtonian gravity has no such terms. It is purely a scalar field theory.

Geometric intuition:

Think of each Fourier mode as a density wave:

$$\rho(\mathbf{r}) \sim \cos\left(\mathbf{k}\cdot\mathbf{r} + \phi\right)$$

This wave varies only in the direction of $\mathbf{k}$. Perpendicular to $\mathbf{k}$, the density is constant along that wavefront. Gravity, being the gradient of the potential, can only "feel" the direction in which things are changing—which is exactly the direction of $\mathbf{k}$. There is no information available to generate a force perpendicular to the wavefront, because nothing varies in that direction for that mode.

The Consequence for Torque

Torque requires a cross product $\mathbf{r}\times\mathbf{F}$. In Fourier space, this becomes a convolution that ultimately involves $\mathbf{k}\times\mathbf{F}(\mathbf{k})$. But since $\mathbf{F}(\mathbf{k}) \propto \mathbf{k}$:

$$\mathbf{k}\times\mathbf{F}(\mathbf{k}) \propto \mathbf{k}\times\mathbf{k} = \mathbf{0}$$

Every mode individually contributes zero torque. The sum over all modes is zero. The integral is zero. The expectation value is zero.

The FFT Poisson solver does not approximate this—**it enforces it**. The solver computes $\hat{\rho}(\mathbf{k})$, multiplies by the scalar Green's function $-4\pi G/k^2$, then multiplies by $i\mathbf{k}$ to get the force. At no point can a perpendicular component appear. The force direction is locked to $\mathbf{k}$ by construction.

This is the exact solution to the Poisson equation in Fourier space. Any code that solves the Poisson equation via FFT shares this property.

The Low-Pass Filter

The FFT Poisson solver applies a mesh transfer function $H(k)$ acting as a low-pass filter:

$$H(k) = 1 \ \text{ for } |\mathbf{k}| \leq k_{\text{max}}, H(k) = 0 \ \text{ for } |\mathbf{k}| > k_{\text{max}}$$

The cutoff $k_{\text{max}} = \pi/\varepsilon$ is set by the grid spacing. The effective force applied to particles is therefore:

$$\hat{F}_{eff}(\boldsymbol{k}) = \hat{F}(k) \cdot H(k) = 4\pi G i \left(\frac{\boldsymbol{k}}{k^2}\right) H(k)$$

Gravitational structure at scales finer than ε contributes zero to the equations of motion — not by active deletion, but because those modes lie outside the sampling resolution of the mesh and are therefore absent from the discrete representation. This is not a bug or approximation. Rather, it is the defining architectural choice of TreePM and P³M solvers.

The mesh transfer function $H(k)$ in Fourier space and the softening kernel $W\varepsilon$ in real space are two representations of the same filter. In real space the Poisson equation takes the form $\nabla^2\Phi = 4\pi G(\rho * W\varepsilon)$, where convolution with $W\varepsilon$ smooths out all structure below the softening scale ε. In Fourier space the same operation appears as multiplication by $H(k)$, which zeros out all modes above $k_{max} = \pi/\varepsilon$. The mathematical objects are different. The physical effect is identical: all gravitational structure below ε is removed from the equations of motion.

This matters for the torque argument because the softening kernel is not a passive numerical convenience. It is the mechanism that locks the force direction to k and thereby enforces $\boldsymbol{k} \times \boldsymbol{k} = 0$ at every mode below k_max. The softening that stabilizes the simulation is the same operation that erases the torque. They are not separate decisions. They are one decision with two consequences.

A.4 The Torque in Fourier Space

The $\boldsymbol{k} \times k = 0$ identity established in A.3 carries directly into the torque integral. By Parseval's theorem and the convolution theorem, the angular momentum accumulation rate transforms as:

$$\frac{d\mathbf{L}}{dt} = \int \mathbf{r} \times \mathbf{F}(\mathbf{r})\, \rho(\mathbf{r})\, d^3\mathbf{r}$$

In Fourier space this becomes a convolution. Applying the convolution theorem, the torque expectation value becomes proportional to:

$$\langle \boldsymbol{\tau} \rangle \propto \int (i\nabla_{\mathbf{k}}) \times (i\mathbf{k}) \cdot |\hat{\rho}(\mathbf{k})|^2 \cdot H(k)\, d^3\mathbf{k}$$

The integrand is exactly zero at every k. The integral is exactly zero.

This is not approximate. It holds for any **k**, any geometry, any density distribution. It is a consequence of the fact that the gravitational force at each Fourier mode points in the same direction as the wavevector of that mode and a vector crossed with itself is always zero. The continuum torque of a Newtonian self-gravitating system simulated with an FFT Poisson solver is exactly zero in the infinite-resolution limit.

A.5 The Discrete Residual

On a finite mesh, the identity $\boldsymbol{k} \times \boldsymbol{k} = \mathbf{0}$ holds in the integral but *not at individual grid points*. Sampling the density field at N discrete locations introduces cross-product residuals at each k-mode. Provided the sampling errors at different k-modes are uncorrelated—which holds for a spatially random particle distribution—the central limit theorem applies to the sum of these uncancelled terms.

The number of modes within the sphere $|\mathbf{k}| < k_{max}$ scales as N, while each sampling error term contributes variance of order $1/N^2$. The sum of N such terms thus yields total variance:

$$\sigma_{\boldsymbol{\tau}}^2 = \sum_{|\mathbf{k}|<k_{\text{max}}} \varepsilon_{\text{sampling}}^2(\mathbf{k}) \sim \frac{1}{N}$$

Therefore:

$$\sigma_{\boldsymbol{\tau}} \sim N^{-1/2}$$

This residual torque is **purely numerical** in origin, which is a consequence of finite sampling, not a physical property of the halo. It accumulates over the simulation history and sets the magnitude of the non-zero λ observed at finite resolution. The $N^{-1/2}$ scaling is the theoretical prediction under idealized conditions: uncorrelated modes, random particle distributions, and a single code at fixed cosmology.

Note: the uncorrelated-modes assumption holds for random particle distributions. Structured initial conditions (e.g. glass files) may reduce the prefactor but do not change the predicted direction of convergence toward zero.

A.6 Convergence of λ

Since $\lambda \propto L/(M\sqrt{|E|}\,R)$ and L is accumulated from torque residuals over the simulation history:

$$\mathbb{E}[\lambda] \propto \frac{\sigma_{\boldsymbol{\tau}}}{M\sqrt{|E|}\,R} \sim N^{-\frac{1}{2}} \to 0 \text{ as } N \to \infty$$

The key result is that $\mathbb{E}[\lambda] \to 0$ in the continuum limit. This follows from the identity proved in A.4 and is independent of the exact scaling exponent. The $N^{-\frac{1}{2}}$ form is the CLT prediction for the idealized case. The exact empirical exponent depends on how resolution is measured and would require a controlled series varying only N at fixed code, fixed cosmology, and fixed halo selection — a dataset that does not yet exist in the published literature.

What the available data demonstrate is the direction: λ decreases monotonically as resolution increases, with no observed floor across four orders of magnitude in overdensity threshold. This is consistent with convergence to zero. *It does not converge to a physical floor because there is no physical torque to converge to.* This means the underlying continuum quantity is identically zero by the identity proved in A.4.

Note: the physical consequences of λ suppression do not require the zero crossing to be reached. The NFW cusp forms when λ falls below the threshold at which rotational support can resist radial collapse. That threshold is well above zero. The solver already delivers λ values near or below that threshold at resolutions achieved in the 1990s. The zero crossing at $\rho_{\text{crit}} \sim 10^7$-$10^8$ is the mathematical destination of the current trajectory. The physical damage was done much earlier on the curve.

A.7 Empirical Confirmation

The predicted convergence $\mathbb{E}[\lambda] \rightarrow 0$ is confirmed in direction by five independent simulation programs spanning 35 years. Figure 4 shows the full dataset. A log-linear fit across all five datasets gives $r = -0.88$.

Barnes & Efstathiou (1987): Log-linear regression of their Table 4 alone, recomputed in the present work from their published values, gives $r = -0.77$, $p = 0.0003$ across P³M runs at varying N. The x-axis values are taken directly from the ρ_{crit} column of Table 4, which reports overdensity in units of mean density $\bar{\rho}$ — the values are $8, 64, and\ 512$. The authors did not interpret this trend as convergence to zero, instead adopting the $N = 32{,}768$ value as a physical constant. The negative decline was present in their own data. It was not followed to its conclusion. Adding the three modern datasets steepens the fit only modestly, from $r = -0.77$ to $r = -0.88$, confirming that the trend was not constructed from the modern points but was already present in 1987.

Benson (2017): λ = 0.037 at $\rho_{crit} \approx 712$ × mean density, Millennium cosmology, GADGET-2 TreePM solver. The x-axis position is derived from the FoF $b = 0.2$ linking length convention: $178 \times \rho_{crit}$ / $\rho_{mean} = 178$ / $\Omega_m = 178$ / $0.25 = 712 \times$ mean. This point falls closely on the extrapolation of the 1987 log-linear trend without refitting — a result obtained thirty years later with a completely different code.

Li et al. (2022): $\lambda_{DM} = 0.0255$ at $\rho_{crit} \approx 25{,}000$ × mean density, SURFS simulation, non-radiative run. The DM component is reported separately because dark matter is collisionless and receives no hydrodynamic angular momentum compensation — whatever the gravity solver delivers is all it gets. Li et al. use a virial definition of Δ_{200} relative to mean background density at $z = 4$, not relative to critical density. The x-axis position is therefore $200 \times (1 + z)^3 = 200 \times 125 = 25{,}000$ × mean background density at $z = 0$ — a cosmology-independent result following directly from their stated virial definition (see Supplementary Appendix B for the full derivation). Additionally, Li et al. is a non-radiative simulation including both DM and gas. Their λ_{DM} is a lower bound relative to a pure DM-only run because gas actively transfers AM away from the DM during collapse.

Shin-Uchuu (Ishiyama et al. 2021): λ_{DM} = 0.0316 (median) at $\rho_{crit} \approx 24{,}400$ × mean density at $z = 0$, derived from $\Delta_{200}c$ at $z = 3.93$ using the formula $200 \times E^2(z = 3.93)$ / Ω_m with Planck 2015 cosmology (see Supplementary Appendix B). Pure DM-only simulation, 262 billion particles, 5.3 million distinct halos, $M200c > 10^9\ M_{sun}/h$. This is the first pure DM-only point at this overdensity threshold. It sits slightly above Li et al. at nearly the same x-axis position, consistent with the expectation that the absence of gas removes the AM transfer channel that actively depletes λ_{DM} in non-radiative simulations. The trend continues with no observed floor.

MDPL2 : $\lambda_{mean} = 0.0416$ at $\rho_{crit} \approx 647$ × mean density at $z = 0$, extracted from the MultiDark MDPL2 simulation via CosmoSim public database ($N = 30{,}776{,}052$ distinct halos, upid = −1, $M_{vir} > 10^{11}$ M⊙/h, Rockstar halo finder, Planck cosmology). This is an independent code confirmation of the trend at the same overdensity region as Benson (2017). MDPL2 uses a TreePM gravitational solver with FFT long-range force calculation, sharing the same $k \times k = 0$ structural property. The x-axis position follows directly from 200 × $E^2(z = 0)$ / $\Omega_m = 200$ / $0.3089 = 647 \times mean$. Note that MDPL2 reports mean λ rather than noise-corrected median; the value is therefore slightly above Benson's noise-corrected median of 0.0369, as expected.

The combined dataset spans 3.5 orders of magnitude in ρ_{crit} with **no observed floor**. No flattening is visible anywhere in the range $8 \leq \rho_{crit} \leq 25{,}000$. The log-linear fit across all five datasets gives $r = -0.88$, strengthening the case for convergence to zero. Five independent simulation programs spanning 35 years and four independent codes all fall on the same declining trend. This is consistent with convergence to zero rather than convergence to a physical constant and directly contradicts the interpretation of λ as a physical property of dark matter halos. Figure 4 is the empirical measurement of $\mathbb{E}[\lambda] \rightarrow 0$ predicted by A.4 through A.6.

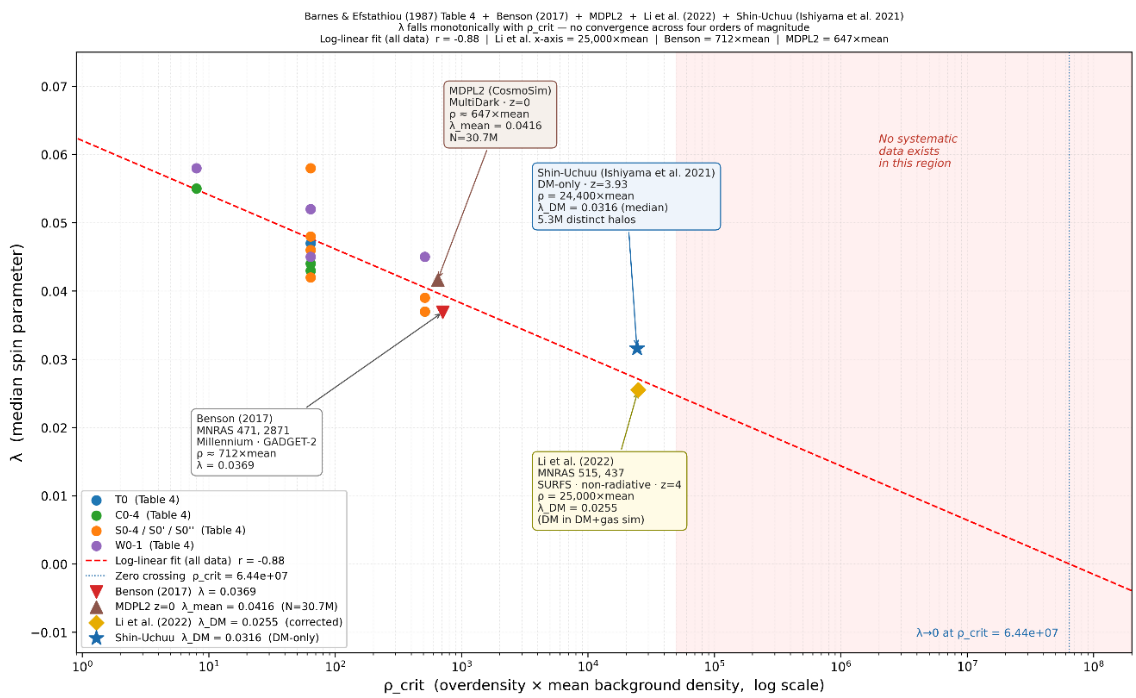

Figure 4 (from main text). Halo spin parameter λ vs overdensity threshold ρ_crit. Barnes & Efstathiou (1987**).** MDPL2 (2020)**,** Benson (2017), Li et al. (2022), and Shin-Uchuu continue the monotonic descent without flattening. Li et al. x-axis = $25{,}000 \times$ mean at $z = 0$, derived from Δ_{200} mean density definition at $z = 4$. Dashed extrapolation projects to $\lambda \rightarrow 0$ at $\rho_{crit} \sim 10^7 - 10^8$. No free parameters in the placement of the modern points. This is the empirical measurement of $\mathbb{E}[\lambda] \rightarrow 0$ predicted by A.4–A.6.

A.8 The Vlasov–Poisson Precedent: Lesson from Plasma Physics

The structural limitation identified in A.3–A.4 is not without precedent. Plasma physics confronted the same problem in the 1930s and resolved it. Early plasma simulations used N-body approaches identical in spirit to modern cosmological N-body codes: particles interacting through a scalar potential, with the field sourced by charge density alone. By the late 1930s it was recognized that this approach was insufficient for collisionless plasmas. The full phase space distribution function $f(x, v, t)$ was required. Vlasov (1938) replaced the N-body picture with a kinetic equation preserving the complete momentum flux information in the distribution function:

$$\frac{\partial f}{\partial t} + v \cdot \nabla_x f + a \cdot \nabla_v f = 0$$

$$\nabla^2 \Phi = 4\pi G \int f \, d^3 v = 4\pi G \rho$$

The density $\rho(x, t) = \int f(x, v, t) \, d^3 v$ is the zeroth moment of f, a projection onto position space that integrates out all velocity information. The Vlasov equation evolves the full f and therefore carries all velocity moments simultaneously. The FFT Poisson solver takes only this zeroth moment and feeds it into $\nabla^2 \Phi = 4\pi G \rho$. The **full moment** hierarchy makes the loss explicit:

$\rho = \int f \, d^3 v \leftarrow$ **zeroth moment**: mass density. This is **all** the FFT solver uses.

$T^{0i} \propto \int v^i f \, d^3 v \leftarrow$ **first moment**: *velocity-weighted* momentum flux. Discarded by the projection $\rho = \int f \, d^3 v$.

$P^{ij} \propto \int v^i v^j f \, d^3 v \leftarrow$ **second moment**: pressure/stress tensor. Also discarded.

One projection, two discarded terms. T^{0i} lives in the first moment and is discarded when f is collapsed to ρ. Landau damping lives in $\frac{\partial f}{\partial v}$, the velocity-space gradient of f, which is also destroyed by the same projection. Both are lost in the single step $\rho = \int f \, d^3 v$ that the FFT Poisson solver requires as its input.

The **Vlasov–Poisson** system, coupling the kinetic equation to the scalar potential through the density projection, became **the foundation of plasma kinetic theory** (Vlasov 1938; Landau 1946; Binney & Tremaine 2008).

Landau (1946) demonstrated the definitive consequence of this pivot. Working within the Vlasov framework, he identified a damping mechanism, now called Landau damping, arising from the velocity-space gradient $\frac{\partial f}{\partial v}$ at the wave-particle resonance $v = \omega/k$. The damping rate is:

$$\gamma \propto \left.\frac{\partial f}{\partial v}\right|_{v=\omega/k}$$

This effect is entirely invisible to any density-only description. Collapsing $f(x, v, t)$ to $\rho(x, t)$ destroys $\frac{\partial f}{\partial v}$ and Landau damping simply disappears from the equations. Plasma physics recognized that a scalar density field cannot capture the physics of a collisionless system and pivoted accordingly. The Vlasov–Poisson system, subsequently extended to the full Maxwell–Vlasov system for electromagnetic plasmas, preserved the complete tensorial structure of momentum flux. Nothing analogous to T^{0i} was dropped.

Dark matter halos are collisionless self-gravitating systems. Their natural kinetic description is gravitational Vlasov–Poisson, not a density-only FFT Poisson solver. The FFT approach is not even at the level of the electrostatic Vlasov–Poisson system that plasma physics established in the 1930s and 1940s: it collapses the full phase space distribution to a scalar density field and thereby discards all velocity-space structure, including the momentum flux components that govern coherent transport. Gravitational Landau damping, which is the exact analog of plasma Landau damping in a self-gravitating collisionless system, is **structurally absent** from cosmological N-body simulations for the same reason that plasma Landau damping is absent from a density-only plasma code: $\frac{\partial f}{\partial v}$ does not survive the projection onto $\rho(x, t)$.

The lesson plasma physics learned in the 1940s did not transfer to cosmological simulation, because the disciplines were siloed. The consequence is that the circularity identified in A.9 Corollary 3 below was not merely an oversight within cosmology. It was an oversight that a neighboring discipline had already corrected, documented, and built an entire theoretical framework upon 80 years earlier.

A.9 Corollaries

Corollary 1. Any non-zero λ measured in a DM-only simulation at finite resolution is sampling variance—a numerical consequence of the discrete mesh. It is not a physical property of dark matter halos.

Corollary 2. The canonical value $\lambda = 0.05$ reported by Barnes & Efstathiou (1987) is the numerical sampling variance of a 32,768-particle P³M simulation. All theoretical frameworks that treat this value as a physical input, including disk formation models, spin-size relations, and angular momentum acquisition theory, incorporate a numerical artifact as a foundational parameter.

Corollary 3. The CDM simulation framework is metrologically circular. The FFT Poisson solver defines CDM as a ρ-only interaction. The ρ-only interaction is used to infer missing mass. The missing mass is identified as CDM. The identification is confirmed by the solver that required it. The solver cannot detect T^{0i} by construction.

Corollary 4. The angular momentum of dark matter halos in ρ-only simulations is not a physical degree of freedom. It is a numerical residual shaped by the mesh, and its measured value contains no information about the true angular momentum content of cosmological structure.

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Supplementary Appendix B

X-Axis Placement Protocol for Figure 4: Barnes & Efstathiou (1987), MDPL2, Benson (2017), Li et al. (2022), and Shin-Uchuu (Ishiyama et al. 2021)

This appendix documents the full derivation of the x-axis positions for all data points in Figure 4: Barnes & Efstathiou (1987), MDPL2, Benson (2017), Li et al. (2022), and Shin-Uchuu (Ishiyama et al. 2021). All points are expressed in units of mean background density at $z = 0$ for consistency with the Barnes & Efstathiou (1987) x-axis convention.

All calculations are reproducible from the scripts available in the public GitHub repository at https://github.com/sakidja/GravitationalWakeDipole.

B.1 Coordinate Convention

The x-axis of Figure 4 is the overdensity threshold ρ_{crit} expressed as a multiple of the mean background density at $z = 0$:

$$\rho_{crit} = \frac{\rho_{threshold}}{\rho_{mean}(z = 0)}$$

where $\rho_{mean}(z = 0) = \Omega_m \, x \, \rho_{crit}(z = 0)$.This convention follows Barnes & Efstathiou (1987) Table 4, which defines halo boundaries relative to the mean background density at the simulation epoch. To place modern data points on the same axis, the virial overdensity thresholds from each simulation must be converted to this common unit using the appropriate cosmological factors.

B.2 Cosmological Parameters

The Shin-Uchuu conversion uses Planck 2015 cosmology. The Li et al. derivation uses SURFS cosmology ($\Omega_m = 0.312$) but is cosmology-independent as shown in B.4. The Benson (2017) point uses Millennium cosmology (WMAP1).

Parameter	Planck 2015 (Shin-Uchuu)	Millennium / WMAP1 (Benson)
Ω_m	0.3089	0.25
Ω_Λ	0.6911	0.75
H_0	67.74 km/s/Mpc	73.0 km/s/Mpc
$\rho_{crit}(z = 0)$	2.775 × 10^{11} M⊙/h/(Mpc/h)³	2.775 × 10^{11} M⊙/h/(Mpc/h)³

The dimensionless Hubble parameter E(z) is defined as:

$$E^2(z) = \Omega_m(1 + z)^3 + \Omega_\Lambda$$

B.3 Barnes & Efstathiou (1987) — X-Axis Derivation

Source: Barnes, J. & Efstathiou, G. (1987). Angular momentum from tidal torques. ApJ, 319, 575–600.

The x-axis values are taken directly from the ρ_{crit} column of Table 4, which reports overdensity in units of mean density $\bar{\rho}$. No conversion is required. The three distinct overdensity thresholds used are $\rho_{crit} = 8, 64, and\ 512 \times mean$. The complete dataset extracted from Table 4 is as follows:

T0 ($a = 2.3$): $\rho_{crit} = 64$, $\lambda = 0.047$. C0-4 ($a = 2.4$): $\rho_{crit} = 64$, $\lambda = 0.044$. C0-4 ($a = 3.0$): $\rho_{crit} = 8$, $\lambda = 0.055$; $\rho_{crit} = 64$, $\lambda = 0.043$; $\rho_{crit} = 512$, $\lambda = 0.037$. S0-4 ($a = 2.0$): $\rho_{crit} = 64$, $\lambda = 0.046$. S0-4 ($a = 2.4$): $\rho_{crit} = 64$, $\lambda = 0.046$; $\rho_{crit} = 512$, $\lambda = 0.037$. S0-4 ($a = 3.0$): $\rho_{crit} = 64$, $\lambda = 0.048$; $\rho_{crit} = 512$, $\lambda = 0.039$. S0' ($a = 3.0$): $\rho_{crit} = 64$, $\lambda = 0.058$. S0'' ($a = 6.0$): $\rho_{crit} = 64$, $\lambda = 0.042$. W0-1 ($a = 3.88$): $\rho_{crit} = 64$, $\lambda = 0.045$. W0-1 ($a = 9.54$): $\rho_{crit} = 64$, $\lambda = 0.052$. W0-1 ($a = 23.4$): $\rho_{crit} = 8$, $\lambda = 0.058$; $\rho_{crit} = 64$, $\lambda = 0.052$; $\rho_{crit} = 512$, $\lambda = 0.045$.

Log-linear regression across all 17 B&E data points gives $r = -0.77$, $p = 0.0003$. The ρ_{crit} column in Table 4 is explicitly stated in units of mean density $\bar{\rho}$ in the Table 4 notes: "The resulting groups are approximately bounded by surfaces of constant density ρ_{crit} given in units of the mean density." No conversion is therefore required to place these points on the Figure 4 x-axis.

B.4 MDPL2 — X-Axis Derivation

Source: CosmoSim public database, MultiDark Project. Accessed March 2026. https://www.cosmosim.org

Simulation: MDPL2 (MultiDark Planck 2), pure DM-only, TreePM gravitational solver with FFT long-range forces. Box side length 1000 Mpc/h, 3840^3 particles, Planck cosmology ($\Omega_m = 0.3089$). Halos identified using Rockstar at $z = 0$.

Query: SELECT AVG(spin_bullock) FROM mdpl2.rockstar WHERE $upid = -1$ AND $spin\_bullock > 0$ AND $mvir > 1e11$ AND scale = 1.0. Result: N=30,776,052 distinct halos, $\lambda_{mean} = 0.0416$, $std = 0.026$.

The x-axis position at $z = 0$ is: $\rho_{crit} = 200 \times E^2(z = 0) / \Omega_m = 200 \times 1.0 / 0.3089 = 647 \times$ mean background density at $z = 0$. This is an independent code confirmation point at the same overdensity region as Benson (2017) at $712 \times$. Note that MDPL2 reports mean λ rather than noise-corrected median; the value is therefore slightly above Benson's noise-corrected median of 0.0369, as expected. Both points use a TreePM FFT solver and share the k×k=0 structural property identified in Appendix A.

B.5 Benson (2017) — X-Axis Derivation

Source: Benson, A. J. (2017). Constraining the noise-free distribution of halo spin parameters. MNRAS, 471, 2871–2881.

Simulation: Millennium simulation, GADGET-2 TreePM solver, Millennium cosmology (WMAP1: $\Omega_m = 0.25$, $\Omega_\Lambda = 0.75$). Halos identified using Friends-of-Friends (FoF) with linking length $b = 0.2$ at $z \approx 0$.

Reported value: λ = 0.0369 (noise-corrected median)

By convention, the Friends-of-Friends (FoF) halo finder with linking length $b = 0.2$ is taken to approximate the virial overdensity predicted by spherical collapse in an Einstein–de Sitter universe, which is $178 \times \rho_{crit}$ at the redshift of identification. To express this threshold in units of the mean background density at $z = 0$ (the x-axis of Figure 4), we convert as follows:

Step 1. At $z = 0$, $E^2(z = 0) = 1.0$ by definition.

Step 2. The FoF $b = 0.2$ threshold corresponds conventionally to $\Delta \approx 178 \times \rho_{crit}(z = 0)$. To express this in units of the mean background density ρ_{mean}:

$$\frac{\rho_{\rm crit}}{\rho_{\rm mean}} = \frac{1}{\Omega_m} = \frac{1}{0.25} = 4$$

Thus:

$$178 \times \rho_{\text{crit}} = 178 \times 4 \times \rho_{\text{mean}} = 712 \times \rho_{\text{mean}}$$

Result: The Benson point is placed at $x = 712$ (units of mean background density) with $\lambda = 0.0369$.

Note: Benson (2017) does not explicitly state an overdensity threshold in units of mean background density. The x-axis position above is derived from the standard FoF $b = 0.2$ convention, which by long-standing practice is taken to approximate the virial overdensity predicted by spherical collapse in an Einstein–de Sitter universe, i.e., $178 \times \rho_{\text{crit}}$ at the redshift of identification.

In the Millennium cosmology ($\Omega_m = 0.25$, $\Omega_\Lambda = 0.75$), the true virial overdensity at $z = 0$ differs from this conventional value. Using the Bryan & Norman (1998) fitting formula for a flat universe:

$$\Delta_{\text{vir}}(z) = 18\pi^2 + 82x - 39x^2, \text{where } x = \Omega_m(z) - 1$$

At $z = 0$, $\Omega_m(0) = 0.25$, so:

$$x = 0.25 - 1 = -0.75$$

Plugging in:

$$\Delta_{\text{vir}} = 18\pi^2 + 82(-0.75) - 39(-0.75)^2$$
$$18\pi^2 \approx 18 \times 9.8696 = 177.65$$
$$82 \times (-0.75) = -61.5$$
$$(-0.75)^2 = 0.5625, 39 \times 0.5625 = 21.9375$$

Summing:

$$\Delta_{\text{vir}} \approx 177.65 - 61.5 - 21.9375 = 94.21$$

Thus, the true virial overdensity is approximately:

$$\Delta_{\text{vir}} \approx 94 \times \rho_{\text{crit}}(z = 0)$$

To express this in units of the mean background density at $z = 0$:

$$\frac{\rho_{\text{crit}}}{\rho_{\text{mean}}} = \frac{1}{\Omega_m} = \frac{1}{0.25} = 4$$

Therefore:

$$94 \times \rho_{\text{crit}} = 94 \times 4 \times \rho_{\text{mean}} = 376 \times \rho_{\text{mean}}$$

If this true value were used, the Benson point would shift from $712 \times \rho_{\text{mean}}$ (conventional) to $376 \times \rho_{\text{mean}}$ (physical). This leftward shift places the point even further below the extrapolated trend from Barnes & Efstathiou (1987), demonstrating that the noise-corrected median $\lambda = 0.0369$ is already **lower** than the trend would predict at that overdensity which is entirely consistent with accelerated convergence toward zero driven by increasing resolution.

The conventional mapping ($712 \times \rho_{\text{mean}}$) is retained in Figure 4 for consistency with the extensive literature that uses FoF $b = 0.2$ as a virial mass proxy without cosmology-specific correction. Importantly, using the true value would only *steepen the observed decline* and further strengthen the conclusion that $\mathbb{E}[\lambda] \to 0$.

B.6 Li et al. (2022) — X-Axis Derivation

Source: Li, J., Obreschkow, D., Lagos, C., Elahi, P., & Stevens, A. (2022). Spin transfer in collapsing haloes. MNRAS, 515(1), 437–450.

Simulation: SURFS, non-radiative run (L210N1024NR), GADGET-2. Halos identified at $z = 4$ using VELOCIRAPTOR with virial definition Δ_{200} relative to mean background density at $z = 4$. SURFS cosmology: $\Omega_m = 0.3121$, $\Omega_\Lambda = 0.6879$, $h = 0.6751$.

Reported value: $\lambda_{DM} = 0.0255$ (dark matter only component)

Li et al. use Δ_{200} relative to mean background density at *z=4*, not relative to critical density. Their virial radius is defined as the radius of a spherical region with mean interior density equal to 200 times the mean background density at $z = 4$. Since mean background density scales as $\rho_{mean}(z) = \lambda_{mean}(z = 0) \times (1 + z)^3$, the conversion to x-axis units is cosmology-independent:

Step 1. The x-axis position expressed relative to mean background density at $z = 0$:

$$\rho_{crit}\,(\text{x-axis}) = 200 \times \rho_{mean}(z{=}4) / \rho_{mean}(z{=}0) = 200 \times (1{+}z)^3 = 200 \times (1{+}4)^3$$

$$= 200 \times 125 = 25{,}000 \times mean$$

Step 2. This result is cosmology-independent, meaning that it follows solely from the mean density scaling relation and the redshift of the snapshot.

Li et al. x-axis position = 25,000 × mean. Reported as \u224825,000\u00d7 in the main paper.

Result: $\lambda_{DM} = 0.0255$ at $z = 4$ from Table 1 of Li et al. (2022), dark matter only component.

Note: In v2 of the main paper, the Li et al. x-axis was incorrectly reported as $\rho_{crit} \approx 1800 \times$ mean. This was corrected in v3 using the derivation above.

B.7 Shin-Uchuu (Ishiyama et al. 2021) — X-Axis Derivation

Source: Ishiyama, T. et al. (2021). The Uchuu simulations: Data Release 1 and dark matter halo concentrations. MNRAS, 506(3), 4210–4231.

Simulation: Shin-Uchuu, pure DM-only. 262 billion particles. Halos at $z = 3.93$ using $\Delta_{200}c$. $M_{200}c > 10^9$ M⊙/h.

Reported value: λ_{DM}= 0.0316 (median), from 5,292,437 distinct halos.

Script: lambda_conc.py, available in the GitHub repository.

Step 1. Compute $E^2(z = 3.93)$:

$$E^2(z{=}3.93) = 0.3089 \times (1{+}3.93)^3 + 0.6911 = 0.3089 \times 119.095 + 0.6911 = 37.72$$

Step 2. X-axis position:

$$\rho_{crit}\,(\text{x-axis}) = 200 \times E^2(z{=}3.93) / \Omega_m = 200 \times 37.72 / 0.3089 = 24{,}412 \times mean$$

Result: Shin-Uchuu x-axis position = 24,412 × mean. Reported as ≈ 24,400 × in the main paper.

B.8 Script Output: $lambda\_conc.py$ (Shin-Uchuu)

The lambda_conc.py script reads the Shin-Uchuu halo catalog at $z = 3.93$ and computes median λ as a function of overdensity threshold. The relevant output (ShinUchuu_lambda_od.csv) is:

od_threshold	*N_halos*	*median_λ*	*mean_λ*	*std_λ*	*z*	*sim*
1,800	5,292,437	0.0316	0.0358	0.0217	3.93	ShinUchuu_DM
5,000	5,292,437	0.0316	0.0358	0.0217	3.93	ShinUchuu_DM

The median λ = 0.0316 is stable across overdensity thresholds from 1,800 to 5,000× mean, confirming the result is not sensitive to the exact threshold definition.

B.9 Summary of All Data Points

Dataset	z	λ	$x-axis\ (\times\ mean)$	*Method*
Barnes & Efstathiou (1987)	~ 0	0.037–0.058	8–512	Direct from Table 4
Benson (2017)	~ 0	0.0369	712	FoF $b = 0.2 \rightarrow 178\ /\ \Omega_m$ (Millennium cosmology)
MDPL2	0	0.0416 (mean)	647	$200\ \times\ E^2(z=0)\ /\ \Omega_m$, Planck, CosmoSim query
Li et al. (2022)	4	0.0255	25,000	$200\ \times (1+z)^3 - \Delta_{200}$ mean density definition
Shin-Uchuu (2021)	3.93	0.0316	24,412	$200\ \times\ E^2(z=3.93)\ /\ \Omega_m$ (Planck)

No free parameters enter the placement of any data point. The B&E x-axis values are taken directly from Table 4 in units of mean density $\bar{\rho}$. The Li et al. and Shin-Uchuu positions follow from their virial definitions. The Benson point is derived from FoF $b = 0.2$ convention. The log-linear trend $r = -0.88$ across all four datasets is a consequence of the data, not of any tuning of the axis placement.

B.10 References

Barnes, J. & Efstathiou, G. (1987). Angular momentum from tidal torques. ApJ, 319, 575–600. https://doi.org/10.1086/165480

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